



BANARAS HINDU UNIVERSITY

B.Sc.(Hons.) Courses

Effective from 2024 under NEP

Offered by

Faculty of Science



BANARAS HINDU UNIVERSITY



Department of Geography

Faculty of Science

SYLLABI

B.Sc. (Hons.) Programme

In

Geography

DEPARTMENT OF GEOGRAPHY
BANARAS HINDU UNIVERSITY, VARANASI-221005

Syllabus

NEP Course Structure for Four Year Undergraduate Geography Program

Semester	Courses	Paper Code	Name of the Paper	Credits
I	Major	GGRMJ11	Physical Geography	3
	Major	GGRMJ12	Map Reading and Interpretation	1
	Minor	GGRMN11	Physical Geography	3
	Minor	GGRMN12	Map Reading and Interpretation	1
	MD1	GGRMD11	Introduction to Earth System	3
	SEC1	GGRSE11	Man and Environment	3
	AEC1	-	MIL	2
	VA C1V AC2	- - -	To be offered by the Institute To be offered by the Institute	2 2
			Total Credits	20

Semester	Courses	Paper Code	Name of the Paper	Credits
II	Major	GGRMJ21	Human Geography	3
	Major	GGRMJ22	Map Projection and Weather Map	1
	Minor	GGRMN21	Human Geography	3
	Minor	GGRMN22	Map Projection and Weather Map	1
	MD2	GGRMD21	Environmental Ethics and Sustainability	3
	SEC2	GGRSE21	Basics of Remote Sensing	3
	AEC2	-	ENG1	2
	VA C3V AC4	- - -	To be offered by the Institute To be offered by the Institute	2 2
			Total Credits	20

Note: Students may have the option to take Offline as well as Online courses. However, students may take up to the 40% of the credits from Online courses in each category. Further, the capping of 40% may be relaxed to 60% for Courses in "Others" category to enable them to Credit more interesting courses. Students will have to pay the Fees for Online Courses.

Semester	Courses	Paper Code	Name of the Paper	Credits
III	Major	GGRMJ31	Economic Geography	3
	Major	GGRMJ32	Surveying	1
	Major	GGRMJ33	Geography of India	4
	Minor Voc1	GGRMV31	Basics of Geographical Information System	4
	MD3	GGRMD31	Global Environmental Challenges	3
	SEC3	GGRSE31	Geography of Tourism	3
	AEC3	-	To be offered by the Institute	2
			Total Credits	20

Semester	Courses	Paper Code	Name of the Paper	Credits
IV	Major	GGRMJ41	Evolution of Geographical Thought	4
	Major	GGRMJ42	Environmental Geography	4
	Major	GGRMJ43	Geomorphology	4
	Major	GGRMJ44	Advanced Map Projection and Geological Map	2
	Minor Voc 2	-	To be offered by the Institute	4
	AEC4	-	To be offered by the Institute	2
			Total Credits	20

Note: Students may have the option to take Offline as well as Online courses. However, students may take up to the 40% of the credits from Online courses in each category. Further, the capping of 40% may be relaxed to 60% for Courses in "Others" category to enable them to Credit more interesting courses. Students will have to pay the Fees for Online Courses.

Semester	Courses	Paper Code	Name of the Paper	Credits
V	Major	GGRMJ51	Climatology	4
	Major	GGRMJ52	World Regional Geography	4
	Major	GGRMJ53	Population Geography	4
	Major	GGRMJ54	Cartographic Techniques	2
	Minor Voc 3	GGRMV51	Hazards & Disaster Risk Reduction	4
	Internship	GGRIN51	Internship1	2
			Total Credits	20

Semester	Courses	Paper Code	Name of the Paper	Credits
VI	Major	GGRMJ61	Advanced Remote Sensing & Geographical Information System	4
	Major	GGRMJ62	Oceanography	4
	Major	GGRMJ63	Remote Sensing and Geographical Information System techniques	2
	Major	GGRMJ64	Quantitative Methods in Geography	2
	Elective (Any one)	GGRMJ65A GGRMJ65B GGRMJ65C GGRMJ65D GGRMJ65E	Hydrology Geography of Settlement Resource Geography Social Geography Cultural Geography	4
	Minor Voc4	GGRMV61	Geography of Health and Wellbeing	4
			Total Credits	20

Note: Students may have the option to take Offline as well as Online courses. However, students may take up to the 40% of the credits from Online courses in each category. Further, the capping of 40% may be relaxed to 60% for Courses in "Others" category to enable them to Credit more interesting courses. Students will have to pay the Fees for Online Courses.

Semester	Courses	Paper Code	Name of the Paper	Credits
VII	Major	GGRMJ71	Research Methodology	4
	Major	GGRMJ72	Environmental Studies	4
	Major	GGRMJ73	Geographical Field Training and Report writing	2
	Elective (Any one)	GGRMJ74A GGRMJ74B GGRMJ74C GGRMJ74D GGRMJ74E GGRMJ74F GGRMJ74G	Advanced Geographical Thought Advances in Population Studies Urban Geography Regional Planning Advanced Geographical Information System Rural and Urban Planning Geography of Migration & Diaspora	4
	Minor	GGRMN71	Research Methodology	4
	Internship	GGRIN71	Internship2	2
			Total Credits	20

Semester	Courses	Paper Code	Name of the Paper	Credits
VIII	Major	GGRMJ81	Advanced Surveying	2
	Major	GGRMJ82	Environmental Quality Analysis	2
	Elective (Any Three)	GGRMJ83A GGRMJ83B GGRMJ83C GGRMJ83D GGRMJ83E GGRMJ83F GGRMJ83G GGRMJ83H GGRMJ83I	Biogeography Advanced Remote Sensing Sensing Advances in Urban Planning Industrial Geography Gender Geography Agriculture Geography Environmental Hazards & Disaster Risk Reduction Ecotourism & Geo heritage Fluvial Geomorphology	12 (4X3)
	Minor	GGRMN81	Advanced Surveying	2
	Minor	GGRMN82	Environmental Quality Analysis	2
			Total Credits	20

Note: Students may have the option to take Offline as well as Online courses. However, students may take up to the 40% of the credits from Online courses in each category. Further, the capping of 40% may be relaxed to 60% for Courses in "Others" category to enable them to Credit more interesting courses. Students will have to pay the Fees for Online Courses.

For Honours Degree with Research

Semester-VII

Semester	Courses	Paper Code	Name of the Paper	Credits
VII	Major	GGRMJ7R1	Research Methodology	4
	Major	GGRMJ7R2	Environmental Studies	4
	Major	GGRMJ7R3	Geographical Field Training & Report writing	2
	Elective (Any one)	GGRMJ7R4A	Advanced Geographical Thought	4
		GGRMJ7R4B	Advances in Population Studies	
		GGRMJ7R4C	Urban Geography	
		GGRMJ7R4D	Regional Planning	
		GGRMJ7R4E	Advanced Geographical Information System	
		GGRMJ7R4F	Rural and Urban Planning	
		GGRMJ7R4G	Geography of Migration & Diaspora	
	Minor	GGRMJ7R4B	Research Methodology	4
	Intern	GGRMJ7R4C	Internship2	2
			Total Credits	20

Semester-VIII

Semester	Courses	Paper Code	Name of the Paper	Credits
VIII	Major	GGRMJ8R1	Advanced Surveying	2
	Major	GGRMJ8R2	Environmental Quality Analysis	2
	Major	GGRDS8R1	Dissertation	12
	Minor	GGRMN8R1	Advanced Surveying	2
	Minor	GGRMN8R2	Environmental Quality Analysis	2
			Total Credits	20

Course	B.Sc./B.A.First Year:Semester-I			
Course Title	GGRMJ11:PhysicalGeography			
Category of Course	Major			
Credits &Hour of Teaching		Theory	Practical	Cumulative
	Credits	3		
	Hour of Teaching (Total)	45		
Course Objectives	- To explain students about the basic concepts of physical geography. - To explain students of physical features with the formation of ocean and land. - To explain students about various theories related to earth's movement. - To acquaint students about atmospheric and hydrological processes.			
Course Content	The course content is divided into 3 Units.			
Units	Course Content			Hr. of Teaching
I	The Earth System: Earth in the solar system; Hypotheses Of the origin of the Earth (Jeans and Jeffreys, Russel) ;Interior of earth. Rocks:origin and classification; Gradational processes; Works of running water, wind, glacier.			15
II	Atmosphere: Composition and structure of atmosphere; Insolation; Temperature: vertical and horizontal distribution; Pressure belts and wind systems, Seasonal and local winds.			15
III	Hydrosphere: Theories of origin of ocean basin (Tetrahedral);Temperature and salinity of ocean water; Ocean currents; Tides; Coral reefs.			15

SEC-Skill Enhancement Course, AEC Ability Enhancement Course; VAC Value Added Course; MD-Multidisciplinary Course 1 Credit (Theory)=15Hours; 1 Credit (Practical)=30Hours

Texts/References	- Barry, R.G. and Chorley, R.J. (1998): Atmosphere, Weather and Climate. Routledge, London. - Bryant, H. R. (2001): Physical Geography Made Simple, Rupa and Company. New Delhi - Bunnett, R. B. (2003): Physical Geography in Diagrams, (4th GCSE edition), Pearson Education (Singapore) Private Ltd., Singapore. - Garrison, T. (1998): Oceanography, Wordsworth Company. Belmont. - Lake, P. (1979): Physical Geography, Cambridge University Press, Cambridge. - Leong, G.C. (2003): Certificate Physical and Human Geography, Oxford University Press, New Delhi. - Monkhouse, F. J. (1979): Physical Geography. Methuen, London - Singh, M. B. (2001): <i>Bhoutik Bhugol</i>, Tara Book Agency, Varanasi - Singh, S. (2003): Physical Geography. Prayag Pustak Bhawan, Allahabad; - Strahler, A.N. and Strahler, A. M. (1992): Modern Physical Geography. John Wiley and Sons, New York. - Trewartha, G. T., Robinson, A.H., Hammond, E. H., and Horn, A. T. (1976/1990): Fundamentals of Physical Geography, (3rd edition), McGraw-Hill, New York. - Holden, J. (2022): Physical Geography: The Basics, (2nd Edition), Routledge, London.
Learning Outcomes	- Students would develop basic concepts of physical geography. - They would get acquainted with physical features of the earth. - They would understand about physical processes of the earth.

Course	B.Sc./B.A.First Year:Semester-I			
Course Title	GGRMJ12:Map Reading and Interpretation			
Category of Course	Major			
Credits & Hour of Teaching		Theory	Practical	Cumulative
	Credits		1	
	Hour of Teaching (Total)		30	
Course Objectives	- To introduce the students to the basics of maps and scales. - To explain the students of various physical features by contours. - To introduce the students to topo sheets. - To acquaint the knowledge in cartography.			
Course Content	The course content is divided into 1 Unit.			
Units	Course Content			Hr. of Teaching
I	Scales and Relief features: Construction of scale: simple, diagonal and comparative; Representation of relief features by contours;			15
II	Profiles and Topographical Map: Profiles (serial, super imposed, projected and composite). Indexing, numbering and scale system, Interpretation of topographical sheet.			15

SEC-Skill Enhancement Course, AEC Ability Enhancement Course; VAC Value Added Course: MD-Multidisciplinary Course 1 Credit(Theory)=15Hours; 1 Credit(Practical)=30Hours

Texts/References	- Monkhouse, F. J. and Wilkinson, F. J. (1972): Maps and Diagrams. Methuen, London. - Raisz, E. (1962): General Cartography, (5th edition), John Wiley and Sons, New York. - Robinson, A. H., Morrison, J. L. and Muehrcke, P. C. (1984): Elements of Cartography, John Wiley and Sons, New York. - Sarkar, A. (2013): Quantitative Geography: : techniques and presentations. Orient Black Swan Private Limited, New Delhi. - Sarkar, A. K. (1997): Practical Geography: A Systematic Approach. Orient Longman, Kolkata. - Sharma, J. P. (2010): Prayogik Bhoogol, Rastogi Publication, Meerut. - Singh, R. L. and Singh, Rana P. B. (1993): Elements of Practical Geography, Kalyani Publishers, New Delhi.
Learning Outcomes	- Students get acquainted about the skills making scales. - Students developed skills to draw relief features with the help of contours and acquaint them about the knowledge of relief features with the help of profiles. - Students get familiar with the knowledge of indexing, numbering and scales of topographical maps. Students get acquainted with the knowledge of topographical maps and their interpretation.

Course	B.Sc./B.A.First Year: Semester-I			
Course Title	GGRMN11:Physical Geography			
Category of Course	Minor			
Credits & Hour of Teaching		Theory	Practical	Cumulative
	Credits	3		
	Hour of Teaching (Total)	45		
Course Objectives	- To explain students about the basic concepts of physical geography. - To explain students of physical features with the formation of ocean and land. - To explain students about various theories related to earth's movement. - To acquaint students about atmospheric and hydrological processes.			
Course Content	The course content is divided into 3 Units.			
Units	Course Content			Hr. of Teaching
I	The Earth System: Earth in the solar system; Hypotheses of the origin of the Earth (Jeans and Jeffreys, Russell); Interior of the earth. Rocks: origin and classification; Gradational processes; Work of running water, wind, glacier			15
II	Atmosphere: Composition and structure of atmosphere; Insolation; Temperature: vertical and horizontal distribution; Pressure belts and wind systems, Seasonal and local winds			15
III	Hydrosphere: Theories of origin of ocean basin (Tetrahedral); Temperature and salinity of ocean water; Ocean currents; Tides; Coral reefs			15

SEC-Skill Enhancement Course, AEC Ability Enhancement Course; VAC Value Added Course: MD-Multidisciplinary Course 1 Credit (Theory)=15 Hours; 1 Credit (Practical)=30 Hours

Texts/References	- Barry, R.G. and Chorley, R.J. (1998): Atmosphere, Weather and Climate. Routledge, London. - Bryant, H. R. (2001): Physical Geography Made Simple, Rupa and Company. New Delhi - Bunnett, R. B. (2003): Physical Geography in Diagrams, (4th GCSE edition), Pearson Education (Singapore) Private Ltd., Singapore. - Garrison, T. (1998): Oceanography, Wordsworth Company. Belmont. - Lake, P. (1979): Physical Geography, Cambridge University Press, Cambridge. - Leong, G.C. (2003): Certificate Physical and Human Geography, Oxford University Press, New Delhi. - Monkhouse, F. J. (1979): Physical Geography. Methuen, London - Singh, M. B. (2001): <i>Bhoutik Bhugol</i>, Tara Book Agency, Varanasi - Singh, S. (2003): Physical Geography. Prayag Pustak Bhawan, Allahabad; - Strahler, A.N. and Strahler, A. M. (1992): Modern Physical Geography. John Wiley and Sons, New York. - Trewartha, G. T., Robinson, A.H., Hammond, E. H., and Horn, A. T. (1976/1990): Fundamentals of Physical Geography, (3rd edition), McGraw-Hill, New York. - Holden, J. (2022): Physical Geography: The Basics, (2nd Edition), Routledge, London.
Learning Outcomes	- Students would develop basic concepts of physical geography. - They would get acquainted with physical features of the earth. - They would understand about physical processes of the earth.

Course	B.Sc./ B.A.First Year:Semester-I			
CourseTitle	GGRMN12:Map Reading and Interpretation			
Category of Course	Minor			
Credits&Hourof Teaching		Theory	Practical	Cumulative
	Credits		1	
	Hour of Teaching (Total)		30	
CourseObjectives	• To introduce the students to the basics of maps and scales. • To explain the students of various physical features by contours. • To introduce the students to topographic sheets. • To acquaint the knowledge in cartography.			
CourseContent	The course content is divided into 1 Unit.			
Units	Course Content			Hr. of Teaching
I	Scales and Relief features: Construction of scale: simple, diagonal and comparative; Representation of Relief features by contours;			15
II	Profiles and Topographical Map: Profiles (serial, superimposed, projected and composite). Indexing, numbering and scale system, Interpretation of topographical sheet.			15

SEC-Skill Enhancement Course, AEC Ability Enhancement Course; VAC Value Added Course: MD-Multidisciplinary Course 1 Credit (Theory)=15Hours; 1Credit(Practical)=30Hours

Texts/References	1. Monkhouse, F.J. and Wilkinson, F.J. (1972): Maps and Diagrams. Methuen, London. 2. Raisz, E. (1962): General Cartography. John Wiley and Sons, New York. 5th edition. 3. Robinson, A., Sale, R. Morrison, J. and Muehrcke, P.C. (1984): Elements of Cartography, 4. Sarkar A. (2013): Quantitative Geography, Orient Black Swan Private Limited 5. Sarkar, A.K. (1997): Practical Geography: A Systematic Approach. Orient Longman Kolkata 6. Sharma, J.P.: Prayogik Bhoogol, Rastogi Publication - Singh, R.L. and Singh, Rana P.B. (1993): Elements of Practical Geography. (English editions).
Learning Outcomes	• Students get acquainted about the skills making scales. • Students developed skills to draw relief features with the help of contours and acquaint them about the knowledge of relief features with the help of profiles. • Student get familiar with the knowledge of indexing, numbering and scales of topographical maps. • Students get acquainted with the knowledge of topographical maps and their interpretation.

Course	B.Sc./B.A.FirstYear:Semester-I			
Course Title	GGRMD11:Introduction to Earth System			
Category of Course	Multi-Disciplinary-1			
Credits & Hour of Teaching		Theory	Practical	Cumulative
	Credits	3		
	Hour of Teaching (Total)	45		
Course Objectives	- To introduce the students to the basics of rotation and revolution. - To explain the students of physical features with the formation of lithosphere. - To introduce the students to various theories related earth's movement - To acquaint the knowledge in biosphere.			
Course Content	The course content is divided into 3 Units.			
Units	Course Content			Hr. of Teaching
I	The Earth System: Earth's Movement; Rotation, Revolution; Interior of the earth. Gradational processes; wind, river and glacier; Types of rocks.			15
II	Atmosphere: Composition and structure; Insolation; Distribution of temperature; Pressure belts and wind systems, seasonal and local winds.			15
III	Hydrosphere: Origin of ocean basin; Temperature and salinity of ocean water; Ocean currents; Coral reefs and tides.			15

SEC-Skill Enhancement Course, AEC Ability Enhancement Course; VAC Value Added Course: MD-Multidisciplinary Course 1 Credit (Theory)=15 Hours; 1 Credit (Practical)=30 Hours

Texts/References	- Barry, R.G. and Chorley, R.J. (1998): Atmosphere, Weather and Climate .Routledge ,London. - Bryant, H. Richard (2001): Physical Geography Made Simple, Rupa and Company. New Delhi - Bunnett, R.B. (2003): Physical Geography in Diagrams, Fourth GCSE edition, Pearson Education (Singapore) Private Ltd. - Garrison, T. (1998): Oceanography, Wordsworth Company. Belmont. - Lake, P. (1979): Physical Geography (English and Hindi editions), Cambridge University Press, Cambridge. - Leong Goh Cheng (2003): Certificate Physical and Human Geography, Oxford University Press, New Delhi. - Monkhouse, F.J. (1979): Physical Geography. Methuen, London Singh, M.B. (2001): <i>Bhoutik Bhugol</i>, Tara Book Agency, Varanasi
Learning Outcomes	- Students would develop understanding of gradational processes. - They would get acquainted with the knowledge of movement of land and ocean. - They would develop an overview about the planetary system.

Course	B.Sc./B.A.FirstYear:Semester-I			
Course Title	GGRSE11:Man and Environment			
Category of Course	SEC-Skill Enhancement Course-1			
Credits & Hour of Teaching		Theory	Practical	Cumulative
	Credits	3		
	Hour of Teaching (Total)	45		
Course Objectives	- To introduce the students to the basics of man and environment relationships. - To explain the students of evolution of man and different biomes. - To introduce the students to forms and functions of ecosystems - To acquaint the knowledge in climate change and environmental conservation			
Course Content	The course content is divided into 3 Units.			
Units	Course Content			Hr. of Teaching
I	Evolution of Man; Human Race: Bases and classification; Human Adaptation in different environment :selected tribes.			15
II	Man's interaction with environment, Biomes: Meaning and types ;Major Biomes: Rainforest, Savannah, Tundra.			15
III	Environment: Meaning and Components ; Food Pyramid; Forms and Functions of Ecosystems: Terrestrial, Aquatic; Biodiversity; Climatic Change; Environmental Conservation and Management.			15

SEC-Skill Enhancement Course, AEC Ability Enhancement Course; VAC Value Added Course: MD-Multidisciplinary Course1 Credit(Theory)=15Hours;1Credit(Practical)=30Hours

Texts/References	- Chisholm, M.(1985):Human Geography, 2nd edition, Penguin Books, London. - deBlij, H.J.(1996):Human Geography: Culture, Society and Space, 2nd edition. John Wiley and Sons, New York. - Johnston, R. J., Gregory, D., Pratt, G. and Watts, M. (2000): The Dictionary of Human Geography. 4th edition, Basil Blackwell Publishers, Oxford. - Kaushik, S.D. and Sharma, A.K.(1996): Principles of Human Geography (in Hindi), Rastogi - Odum, P.E. and Barret, W.G.(2005): Fundamentals of Ecology, Thomson Asia Pvt Ltd, Singapur. - Huggett, R.J.(1998): Fundamentals of Biogeography, Routledge, London. - Saxena, H.M.(2000): Environmental Management. Rawat Publications., Jaipur and New Delhi.
Learning Outcomes	- Students would get acquainted with knowledge of man and environment relationships. - They would get acquainted with knowledge of human evolution and adaptations. - They would develop knowledge of concepts of climatic change; environmental conservation and management.

Course	B.Sc./B.A.FirstYear:Semester-II			
CourseTitle	GGRMJ21:Human Geography			
Category of Course	Major			
Credits&Hourof Teaching		Theory	Practical	Cumulative
	Credits	3		
	Hour of Teaching (Total)	45		
CourseObjectives	- To introduce the nature, scope and approaches of human geography. - To explain the student's human adaptation in environment. - To introduce the student to various theories of population. - To acquaint the knowledge in urban settlement and their growth patterns.			
CourseContent	The course content is divided into 3 Units.			
Units	Course Content			Hr.of Teaching
I	Human Geography: Nature, scope and Approaches of human geography ;Schools of thought in human geography :determinism, possibilism, neo-determinism and Probabilism.			15
II	Human Race and Society: Evolution and Classification of races; Characteristics of races and their broad distribution; Human adaptation to environment; Eskimo, Masai, Bushman; Indian tribes: Tharu, Naga, Bhill.			15
III	Population and Settlements: Population Growth and Major human agglomerations; Demographic Transition Theory; Migration: Types, Causes and Consequences; Rural settlement: characteristics, types and patterns in India :Urban settlements: evolution and classification.			15

SEC-Skill Enhancement Course, AEC Ability Enhancement Course; VAC Value Added Course; MD-Multidisciplinary Course 1 Credit (Theory)=15 Hours; 1 Credit (Practical)=30 Hours

Texts/References	- Anderson, K. (2006): Races and crises of Human Development, Routledge, London and New Delhi - Bhende, A.A. and Kanetkar, T. (1978): Principles of Population Studies, Himalayan Publishing House, Mumbai. - Chisholm, M. (1985): Human Geography, 2nd edition, Penguin Books, London. - Fellman, J.D., Arthur, G., Judith, G., Hopkins, J. and Dan, S. (2007): Human Geography: Landscapes of Human Activities. MNGraw-Hill, New York. 10th edition. - Haggett, P. (2004): Geography: A Modern Synthesis. 8th edition, Harper and Row, New York - Hussain, M. (1994): Human Geography, Rawat Publications, Jaipur. - Johnston, R.J., Gregory, D., Pratt, G. and Watts, M. (2009): The Dictionary of Human Geography. 5th edition, Basil Blackwell Publishers, Oxford. - Kaushik, S.D. and Sharma, A.K. (1996): Principles of Human Geography (in Hindi), Rastogi Publication, Meerut. - Norton, W. (2008): Human Geography, Oxford University Press, New York. 5th ed. - Smith, D.M. (1977): Human Geography-A Welfare Approach, Edward Arnold (Publishers) Ltd., London. - Stoddard, R.H., Wishart, D.J. and Blouet, B.W. (1986): Human Geography. Prentice-Hall, Englewood Cliffs, New Jersey.
Learning Outcomes	- Students would develop understanding about the meaning and field of human geography. - They would get acquainted with the knowledge of evolution and classification of human races. - They would understand the theories of population and migration. - They would become aware about rural and urban settlements.

Course	B.Sc./B.A.FirstYear:Semester-II			
CourseTitle	GGRMJ22:Map Projection and Weather Map			
Category of Course	Major			
Credits&Hourof Teaching		Theory	Practical	Cumulative
	Credits		1	
	Hour of Teaching (Total)		30	
CourseObjectives	• Tointroducethestudentsabouttheearthrepresentation. • Toexplainthestudentsaboutclassificationofmapprojections. • Tointroducethestudentsaboutintroductiontoweathermaps.			
CourseContent	Thecoursecontentisdividedinto1Unit.			
Units	CourseContent			Hr.of Teaching
I	Map Projection (Perspective Projection): Introduction and Classification of Map projections; Cylindrical: equal area; Bonne's; Zenithal Projection: Gnomonic, Stereographic and Orthographic			15
II	Weather Map: Introduction to weather map, Conventional signs and symbols; Interpretation of Indian daily weather map: July, October and January months			15

SEC-Skill Enhancement Course, AEC Ability Enhancement Course; VAC Value Added Course: MD-Multidisciplinary Course 1 Credit (Theory)=15 Hours; 1 Credit (Practical)=30 Hours

Texts/References	1. Hinks, A. R. (1921): Map Projection, Cambridge University Press, London, 2. Misra, R. P. and Ramesh A. (1989): Fundamentals of Cartography, Concept Publishing Company, New Delhi. 3. Monkhouse, F. J. and Wilkinson, F. J. (1985): Maps and Diagrams. Methuen, London. 4. Raisz, E. (1962): Principles of Cartography, McGraw Hill, New York. 5. Robinson, A. H., Sale, R., Morrison, J. and Muehrcke, P. C. (1984): Elements of Cartography, 5th edition, John Wiley and Sons, New York, 6. Sarkar, A. K. (1997): Practical Geography: A Systematic Approach, Orient Longman, Kolkata, 7. Sharma, J. P. (2001): Prayogik Bhugol, Rastogi Publication, Meerut 3rd edition, 8. Singh, R. L. and Singh, Rana P. B. (1993): Elements of Practical Geography, (Hindi and English edition), Kalyani Publishers, New Delhi, - Steers, J. A. (1965): An Introduction to the Study of Map Projection, University of London Press, London.
Learning Outcomes	• Students would get knowledge the knowledge of principles and types of map projections. • Student will be able to understand weather symbols. • Students would get acquainted with the knowledge of representation of atmospheric features on maps.

Course	B.Sc./B.A.First Year:Semester-II			
CourseTitle	GGRMN21:Human Geography			
Category of Course	Minor			
Credits&Hour of Teaching		Theory	Practical	Cumulative
	Credits	3		
	Hour of Teaching (Total)	45		
CourseObjectives	- To introduce the nature, scope and approaches of human geography. - To explain the students about human adaptation to environment. - To introduce the students to various theories of population. - To acquaint the knowledge in urban settlement and their growth patterns.			
CourseContent	The course content is divided into 3 Units.			
Units	CourseContent			Hr. of Teaching
I	Human Geography: Nature, scope and Approaches of human geography; Schools of thought in human geography: determinism, possibilism, neo-determinism and Probabilism.			15
II	Human Race and Society: Evolution and Classification of races; Characteristics of races and their broad distribution; Human adaptation to environment; Eskimo, Masai, Bushman; Indian tribes: Tharu, Naga, Bhil.			15
III	Population and Settlement: Population Growth and Major human agglomerations; Demographic Transition Theory; Migration: Types, Causes and Consequences; Rural settlement: characteristics, types and patterns in India: Urban settlements: evolution and classification.			15

SEC-Skill Enhancement Course, AEC Ability Enhancement Course; VAC Value Added Course: MD-Multidisciplinary Course 1 Credit (Theory)=15 Hours; 1 Credit (Practical)=30 Hours

Texts/References	- Anderson, K. (2006): Races and crises of Human Development, Routledge, London and New Delhi - Bhende, A. A. and Kanetkar, T. (1978): Principles of Population Studies, Himalayan Publishing House, Mumbai. - Chisholm, M. (1985): Human Geography, 2nd edition, Penguin Books, London. - Fellman, J. D., Arthur, G., Judith, G., Hopkins, J. and Dan, S. (2007): Human Geography: Landscapes of Human Activities. MNGraw-Hill, New York. 10th edition. - Haggett, P. (2004): Geography: A Modern Synthesis. 8th edition, Harper and Row, New York - Hussain, M. (1994): Human Geography, Rawat Publications, Jaipur. - Johnston, R. J., Gregory, D., Pratt, G. and Watts, M. (2009): The Dictionary of Human Geography. 5th edition, Basil Blackwell Publishers, Oxford. - Kaushik, S. D. and Sharma, A. K. (1996): Principles of Human Geography (in Hindi), Rastogi Publication, Meerut. - Norton, W. (2008): Human Geography, Oxford University Press, New York. 5th ed. - Stoddard, R. H., Wishart, D. J. and Blouet, B. W. (1986): Human Geography. Prentice-Hall, Englewood Cliffs, New Jersey.
Learning Outcomes	- Students would develop understanding about the meaning and field of human geography. - They would get acquainted with the knowledge of evolution and classification of human races. - They would understand the theories of population and migration. - They would become aware about rural and urban settlements.

Course	B.Sc./B.A.First Year:Semester-II			
Course Title	GGRMN22:Map Projection and Weather Map			
Category of Course	Major			
Credits&Hour of Teaching		Theory	Practical	Cumulative
	Credits		1	
	Hour of Teaching (Total)		30	
Course Objectives	- To introduce the students about the earth representation. - To explain the students about classification of map projections - To introduce the students to Introduction to weather map			
Course Content	The course content is divided into 1 Units.			
Units	Course Content			Hr. of Teaching
I	Map Projection (Graphical): Introduction and Classification of Map projections; Cylindrical: equal area; Bonne's; Zenithal Projection: Gnomonic, Stereographic and Orthographic.			15
II	Weather Map: Introduction to weather map, Conventional sign and symbols; Interpretation of Indian daily weather map: July, October and January months.			15

SEC-Skill Enhancement Course, AEC Ability Enhancement Course; VAC Value Added Course: MD-Multidisciplinary Course 1 Credit (Theory)=15 Hours; 1 Credit (Practical)=30 Hours

Texts/References	- Hinks, A. R. (1921): Map Projection, Cambridge University Press, London, - Misra, R. and Ramesh A. (1989): Fundamentals of Cartography, Concept Publishing Company, New Delhi. - Monkhouse, F. J. and Wilkinson, F. J. (1985): Maps and Diagrams. Methuen, London. - Raise, E. (1962): Principles of Cartography, M. N. Graw Hill, New York. - Robinson, A. H., Sale, R., Morrison, J. and Muehrcke, P. C. (1984): Elements of Cartography, 5th edition, John Wiley and Sons, New York, - Sarkar, A. K. (1997): Practical Geography: A Systematic Approach, Orient Longman, Kolkata - Sharma, J. P. (2001): Prayogik Bhugol, Rastogi Publication, Meerut 3rd edition, - Singh, R. L. and Singh, Rana P. B. (1993): Elements of Practical Geography, (Hindi and English edition), Kalyani Publishers, New Delhi, - Steers, J. A. (1965): An Introduction to the Study of Map Projection, University of London Press, London.
Learning Outcomes	- Students would get knowledge of principles and types of map projections. - Student would develop knowledge of weather symbols. - Students would get acquainted with the knowledge of representation of atmospheric features on maps.

Course	B.Sc./B.A.FirstYear:Semester-II			
CourseTitle	GGRMD21:Environmental Ethics and Sustainability			
Category of Course	Multi-Disciplinary-2			
Credits&Hourof Teaching		Theory	Practical	Cumulative
	Credits	3		
	Hour of Teaching (Total)	45		
CourseObjectives	- To introduce the students to the basics of environmental ethics. - To explain the students of various theories. - To introduce the students of green development and biodiversity - To acquaint the knowledge about sustainability: concept and types; major approaches.			
CourseContent	The course content is divided into 3 Units.			
Units	CourseContent			Hr.of Teaching
I	Environmental Ethics: Concept and meaning; science, ethics and environment; Moral theories- anthropocentrism, biocentrism, ecocentrism, deep ecology; Population growth and resource linkages; Ecological niche.			15
II	Decolonial Environmentalism; Idea of green development; Biodiversity and natural capital assessment; Environmental justice and activism; Environmental legislations in India; Indian traditional knowledge system and ecological values;			15
III	Sustainability: Concept and types; Major approaches; Sustainable habitat; Biodiversity challenges and restoration; Regional threats of climate change and mitigation; Geopolitics of food production; Principles of sustainability; Implications and impacts of Sustainable Development Goals.			15

SEC-Skill Enhancement Course, AEC Ability Enhancement Course; VAC Value Added Course; MD-Multidisciplinary Course 1 Credit (Theory)=15 Hours; 1 Credit (Practical)=30 Hours

Texts/References	- Ahlawat, Ajay (2019). Sustainable Development Goals-Directive Principles for Sustainable India by 2030, Notion Press, New Delhi - Edwards, Andres R. and David W. Orr (2005). The Sustainability Revolution: Portrait of a Paradigm Shift, New Society Publishers, Canada - Tomar, Anjani Singh and Amit Kumar Kashyap (2021). Sustainable Development: India and Beyond, Bharti Publications, New Delhi - Smith, Kimberly K. (2018). Exploring Environmental Ethics: An Introduction, Springer Cham - Tiwari, Devendra Nath and Ananda Msihra (2012). Environmental Ethics: Indian Perspectives, Banaras Hindu University, Varanasi - Zagonari, Fabio (2022). Environmental Ethics, Sustainability and Decisions- Literature Problems and Suggested Solutions, Springer Cham
Learning Outcomes	- Students would inculcate the value of environmental ethics. - Students would get acquainted with the knowledge of environmentalism. - Student would understand concepts of biodiversity challenges and restoration. - Students would develop awareness about environmental threats and gain knowledge about Sustainable Development Goals..

Course	B.Sc./B.A.First Year:Semester-II			
CourseTitle	GGRSE21:Basics of Remote Sensing			
Category of Course	SEC-Skill Enhancement Course-2			
Credits&Hour of Teaching		Theory	Practical	Cumulative
	Credits	3		
	Hour of Teaching (Total)	45		
CourseObjectives	- To introducethestudentstothebasicsofremotesensing. - ToexplainthestudentsofSpectralregionsandbands. - TointroducethestudentsaboutRemoteSensing satellites: Platformsandsensors - ToacquainttheknowledgeinRemoteSensingapplicationinresourcemapping.			
CourseContent	Thecoursecontentisdividedinto3Units.			
Units	Course Content			Hr.of Teaching
I	RemoteSensing: ConceptandScope; Electro-magnetic Radiation: Characteristics, Spectral regions and Bands; Interaction with earth surface features and atmosphere; Spectral Signature			15
II	Types of Remote Sensing: Air borne and Space borne; Aerial photos:Types and Characteristics;RemoteSensing satellites: Platforms and sensors			15
III	Visual and Digital image processing techniques: Remote Sensing application in resource mapping and environmental monitoring			15

SEC-Skill Enhancement Course, AEC Ability Enhancement Course; VAC Value Added Course: MD-Multidisciplinary Course 1 Credit (Theory)=15 Hours; 1 Credit (Practical)=30 Hours

Texts/References	- Curran, P.J. (1985): Principles of Remote Sensing, Longman, London - Lillesand, T.M. and Kiefer, R.W. (2000): Remote Sensing and Image Interpretation. 4th edition. John Wiley and Sons, New York - Campbell, J.B. (2002): Introduction to Remote Sensing, 5th edition, Taylor and Francis, London - Bhatta, B. (2010): Remote Sensing and GIS, Oxford University Press, New Delhi. - Nag Prithvi and Kudrat M. (1998): Digital Remote Sensing, Concept Publishing Company, New Delhi
Learning Outcomes	- Students would inculcate the value of environmental ethics. - Students would get acquainted with the knowledge of environmentalism. - Student would understand concepts of biodiversity challenges and restoration. - Students would develop awareness about environmental threats and gain knowledge about Sustainable Development Goals..

Course	B.Sc./B.A.SecondYear:Semester-III			
Course Title	GGRMJ31:Economic Geography			
Category of Course	Major			
Credits &Hour of Teaching		Theory	Practical	Cumulative
	Credits	3		
	Hour of Teaching (Total)	45		
Course Objectives	- To introduce the students to the basics of economic geography. - To explain the students to the location of economic activities. - To introduce the students about agricultural regions and types of industries. - To acquaint the knowledge about world transportation and SEZs.			
Course Content	The course content is divided into 3 Units.			
Units	Course Content			Hr.of Teaching
I	Conceptual Aspects: Meaning, concept and scope of Economic Geography; Classification of economic activity; Locational Theories: Agriculture (Von Thunen), Industry (Weber)			15
II	Primary activities: Agriculture, Fishing, Forestry, and mining ; Agricultural Region of D. Whittlesey; Secondary activities: Types and locational factors of industries ; Iron and Steel and Cotton textiles industry; Industrial regions of World;			15
III	Tertiary and other specialized activities: World transportation: Sea routes and major trans-continental railways; International trade and World Trade Organization (WTO); patterns and trends; Special Economic Zones (SEZs) and Technology Parks			15

SEC-Skill Enhancement Course, AEC Ability Enhancement Course; VAC Value Added Course: MD-Multidisciplinary Course 1 Credit (Theory)=15 Hours; 1 Credit (Practical)=30 Hours

Texts/References	- Alexander, J.W. (1988): Economic Geography. Prentice-Hall, New Delhi - Bagchi-Sen S. and Smith H.L., 2006: Economic Geography: Past, Present and Future, Taylor and Francis. - Bryson, J., Henry, N., Keeble, D. and Martin, R. (eds.) (1999): The Economic Geography Reader: Producing and Consuming Global Capitalism. John Wiley and Sons, Inc, New York. - Clark, G.L., Gertler, M.S. and Feldman, M.P. (eds.) (2000): The Oxford Handbook of Economic Geography. Oxford University Press, USA. - Coe, N. (2007): Economic Geography: A Contemporary Introduction. Blackwell Publishers, Inc., Massachusetts. - Hanink, D.M. (1997): Principles and Applications of Economic Geography: Economy, Policy, Environment. John Wiley and Sons, Inc, New York. - Hartshorne, T.A. and Alexander, J.W. (1988): Economic Geography (3rd revised edition) Englewood Cliff, New Jersey, Prentice Hall - Hodder B.W. and Lee Roger, 1974: Economic Geography, Taylor and Francis
Learning Outcomes	- Students would understand the basics of economic geography. - Acquaint the knowledge of the location of primary and secondary activities. - Students understand the agricultural and industrial regions. - Students would be aware of tertiary activities and SEZs.

Course	B.Sc./B.A.SecondYear:Semester-III			
CourseTitle	GGRMJ32:Surveying			
Category of Course	Major			
Credits&Hour of Teaching		Theory	Practical	Cumulative
	Credits		1	
	Hour of Teaching (Total)		30	
CourseObjectives	• To introducethestudentstothebasics ofsurveyingmethod. • ToexplainthestudentstoChainTapeSurveying;PlaneTableSurveying • To introducethestudentsaboutIndianClinometerandPrismaticCompass surveying. • ToacquainttheknowledgeaboutModernSurveyingTechniques			
CourseContent	Thecoursecontentisdividedinto1Unit.			
Units	CourseContent			Hr.of Teaching
I	BasicsofSurveying: Meaningandscope;Chain& Tape,PlainTable ;IndianClinometer			15
II	ModernSurveying: Prismatic Compass,GPS			15

SEC-SkillEnhancementCourse, AECAbilityEnhancementCourse; VACValueAdded
Course: MD-MultidisciplinaryCourse1Credit(Theory)=15Hours;1Credit(Practical)=30Hours

Texts/References	1. Monkhouse,F.J.&Wilkinson,F.J.(1985)MapsandDiagrams.Methuen,London 2. Raisz,E(1962)GeneralCartography.JohnWiley&Sons,NewYork. 3. Sharma,J.P.(2001)PrayogikBhoogol.RastogiPub,Meerut. 4. SinghR.L.&Singh,RanaPB(1993)ElementsofPracticalGeography(Hindi&EnglishEdition), Kalyani Publishers,NewDelhi. 5. Singh,LR(2006)FundamentalsofPracticalGeography.ShardaPustakBhawan,Allahabad.
Learning Outcomes	• Studentswouldgetknowledgeofthebasics ofsurveyingmethods. • AcquainttheknowledgeofIndianclinometer. • Studentswouldunderstandthe Prismatic Compasssurveying. • To make students aware of the use and applications of GPS.

Course	B.Sc./B.A.SecondYear:Semester-III			
CourseTitle	GGRMJ33:Geography of India			
Category of Course	Major			
Credits&Hour of Teaching		Theory	Practical	Cumulative
	Credits	4		
	Hour of Teaching (Total)	60		
CourseObjectives	- To introduce the student to the basics of physiography and resources. - To explain the student to about agriculture system of India. - To introduce the student about industry, transport and trade system of India To acquaint the knowledge about major geographical regions.			
CourseContent	The course content is divided into 4 Units.			
Units	Course Content			Hr. of Teaching
I	Physiography and Resources: Geology; Physiographic division; Drainage systems; Climate and climatic regions; Soils and natural vegetation; Mineral and power resources: iron ore and coal.			15
II	Agriculture: Means of irrigation; Major crops: rice, wheat, cotton, sugarcane and tea; Agricultural regions; Green Revolution and its consequences; Multipurpose projects: Damodar valley and Bhakra Nangal.			15
III	Industry, Transport, and Trade: Major industries: iron and steel, cotton textile and sugar; Transport network and trade: structure and recent trends; Special Economic Zones (SEZs) and Exclusive Economic Zones (EEZs)			15
IV	Regionalization: Geographical Region: R.L.Singh; Economic Region: P.Sen Gupta and Cultural Region: N.K.Bose ;Regional studies: Garhwal Himalaya Region, Middle Ganga Plain and Indian Desert.			15

SEC-Skill Enhancement Course, AEC Ability Enhancement Course; VAC Value Added Course: MD-Multidisciplinary Course 1 Credit (Theory)=15Hours; 1Credit(Practical)=30Hours

Texts/References	54 Chauhan, P.R. and Prasad, M. (2003): Bharat Ka Vrihad Bhugol, Vasundhara Prakashan, Gorakhpur. - Gautam, A. (2006): Advanced Geography of India, Sharda Pustak Bhawan, Allahabad - Khullar, D.R. (2007): India: A Comprehensive Geography, Kalyani Publishers, New Delhi - Nag, P. and Gupta, S.S. (1992): Geography of India, Concept Publishing Company, New Delhi. - Rao, B.P. (2007): Bharat kee Bhaugolik Sameeksha, Vasundhara Prakashan, Gorakhpur. - Sharma, T.C. and Coutinho, O. (2003): Economic and Commercial Geography of India, Vikas Publishing House Private Ltd. New Delhi. - Singh, J. (2003): India: A Comprehensive Systematic Geography. Gyanodaya Prakashan, Gorakhpur - Singh, J. (2001): Bharat: Bhaugolik Aadhar Avam Ayam, Gyanodaya Prakashan, Gorakhpur. - Singh, R.L. (ed.) (1971): India: A Regional Geography. National Geographical Society of India, Varanasi. - Spate, O.H.K., Learmonth A.T.A. and Farmer, B.H. (1996): India, Pakistan and Sri Lanka. Methuen, London, 7th edition. - Tiwari, R.C. (2007): Geography of India, Prayag Pustak Bhawan, Allahabad.
Learning Outcomes	- Students would get knowledge of the physiography of India. - Acquaint the knowledge of Indian climate and vegetation. - Students will be aware of the agricultural system of India - Students should understand the industry, transport, trade and regions.

Course	B.Sc./B.A.SecondYear:Semester-III			
Course Title	GGRMV31:Basics of Geographical Information System			
Category of Course	Minor Voc			
Credits & Hour of Teaching		Theory	Practical	Cumulative
	Credits	4		
	Hour of Teaching (Total)	60		
Course Objectives	- To introduce the students to the basics of GIS Mapping. - To explain the students about scanning, digitization & editing. - To introduce the students about georeferencing. - To acquaint the knowledge of cartographic conventions.			
Course Content	The course content is divided into 4 Units.			
Units	Course Content			Hr.of Teaching
I	Fundamentals of Geographical Information System: Definition, Evolution of GIS, Components of Geographical Information System; Recent advances in GIS; Coordinate systems, Datums and projections in GIS; GIS Software (Open Source and commercial)			15
II	Geographical Information System Data: Raster and Vector data models; Data Base Management System (DBMS); Data Sources; Data Quality: meaning, measures, components and management; Geographical Information System Data Standards: meaning, classification, and components.			15
III	Geographical Information System Data Analysis: Raster Data Analysis: Local, neighbourhood and regional operations- overlay analysis, map algebra; Vector Data Analysis: Concept of Topology; Proximity Analysis: Buffering, Thiessen Polygon; Essentials of Map Composition.			15
IV	Applications of Geographical Information System: Land use/Land Cover Mapping, geological and geomorphological; Water Resource Management, Disaster Management and Mitigation			15

SEC-Skill Enhancement Course, AEC Ability Enhancement Course; VAC Value Added Course: MD-Multidisciplinary Course 1 Credit (Theory)=15 Hours; 1 Credit (Practical)=30 Hours

Texts/References	1. Bhatia, J.B. (2008) Remote Sensing & GIS, Oxford. 2. Bonham, Carter G.F. (1995): Information Systems for Geoscientists-Modelling with GIS. Pergamon, Oxford. 3. Burrough, P.A. (1986) Principles of GIS for Land Resource Assessment, Oxford. 4. Burrough, P.A. and MNDonnell, R. (1998): Principles of Geographic Information Systems. Oxford University Press, Oxford. London 5. Chang, K.T. (2003): Introduction to Geographic Information Systems. Tata MNGraw Hill Publications Company, New Delhi. 6. Chauniyal, D.D. (2004): Remote Sensing and Geographic Information Systems. (in Hindi) .Sharda Pustak Bhawan, Allahabad. 7. ESRI (1993): Understanding GIS. Redlands, USA 8. Fraser Taylor, D.R. (1991): Geographic Information Systems. Pergamon Press, Oxford. 9. Goodchild, M.F.; Park, B. O. and Steyaert, L. T. (eds.) (1993): Environmental Modelling with GIS. Oxford University Press, Oxford. 10. Guptill, S.C., and Morrison, J.L. (1995): Elements of Spatial Data Quality. Elsevier/Pergamon, Oxford.
Learning Outcomes	- Students would get the concept of GIS. - Students would be aware of types of data models and DBMS - They would be able to understand the raster and vector data analysis. - Students would understand the applications of Geographical Information Systems.

Course	B.Sc./B.A.SecondYear:Semester-III			
CourseTitle	GGRMD31:Global Environmental Challenges			
Category of Course	Multi-Disciplinary-III			
Credits&Hourof Teaching		Theory	Practical	Cumulative
	Credits	3		
	Hour of Teaching (Total)	45		
CourseObjectives	• To introduce the students about biodiversity loss and degradation, • To explain the students about Food and water insecurity; energy crisis, • To introduce the students about climate change. • To acquaint the knowledge of ocean changes and its global impacts.			
CourseContent	The course content is divided into 4 Units.			
Units	CourseContent			Hr.of Teaching
I	Land Related Issues: Population explosion, Biodiversity loss, Land degradation and desertification, Soil and water degradation, Solid waste, Food and water insecurity; Energy crisis; Forced migration; Infectious disease and pandemic.			15
II	Atmosphere Related Issues: Air Pollution, Green house gases and ozone depletion, Atomic radiation, Global warming, Flood and famine. Climate Change.			15
III	Ocean Related Issues: Melting ice caps and sea level rise, Ocean acidification, Overfishing, Commercial whaling, Protecting ocean sanctuaries, Conflicts on territorial water.			15

SEC-Skill Enhancement Course, AEC Ability Enhancement Course; VAC Value Added Course: MD-Multidisciplinary Course 1 Credit (Theory)=15 Hours; 1 Credit (Practical)=30 Hours

Texts/References	1. Giacomo Cao, Roberto Orrù (2014): Current Environmental Issues and Challenges, Springer Nature. 2. Mal, Suraj (eds.) (2017): Climate Change, Extreme Events and Disaster Risk Reduction, Springer Nature. 3. Nakanishi, Yumiko (2016): Contemporary Issues in Environmental Law, Springer Nature. 4. Peszko, Grzegorz and et.al. (2023): Air Pollution and Climate Change, Washington, DC: World Bank Group. 5. Sharma, Rahul (2019): Environmental Issues of Deep-Sea Mining, Springer Nature. 6. Uimonen, Peter and Whalley, John (1997): Environmental Issues in the New World Trading System, Springer Nature.
Learning Outcomes	• Students would get knowledge of biodiversity loss and soil and water degradation. • Students would be aware of global warming and climate change. • Students would be able to understand the ocean acidification. • Students would understand the conflicts on territorial water.

Course	B.Sc./B.A.SecondYear:Semester-III			
CourseTitle	GGRSE31:Geography of Tourism			
Category of Course	SEC-SkillEnhancementCourse-III			
Credits&Hourof Teaching		Theory	Practical	Cumulative
	Credits	3		
	Hour of Teaching (Total)	45		
CourseObjectives	- To introduce the students on nature, and scope of Geography of Tourism. - To explain the students about Tourism as an industry. - To introduce the students about Models of Ecotourism; Impacts of Tourism - To acquaint the knowledge of Sustainable Tourism Development Planning.			
CourseContent	The course content is divided into 3 Units.			
Units	CourseContent			Hr.of Teaching
I	Conceptual: Nature, and Scope of Geography of Tourism; Data sources; Typology of Tourism: developed and developing world; Tourism as an industry; Drivers of Tourism; Tourism and Regional Economy; Concept of heritage and its types.			15
II	System of Tourism and Impacts: Tourism Multiplier and Forecasting Methods; Trends and Pattern of tourism in the World; Models of Ecotourism; Impacts of Tourism- Socio-cultural problems and Environmental degradation.			15
III	Policy and Planning for Heritage-Tourism in India: Of Cultural Heritage: Conservation and Revitalization; Legislation and Acts; Sustainable Tourism Development Planning; Role of tourism in development.			15

SEC-Skill Enhancement Course, AEC Ability Enhancement Course; VAC Value Added Course: MD-Multidisciplinary Course 1 Credit (Theory)=15 Hours; 1 Credit (Practical)=30 Hours

Texts/References	1. Bhatia A.K. (1996): Tourism Development: Principles and Practices, Sterling Publishers, New 2. Carter, E and G. Lowman (1994): Ecotourism, John Wiley and Sons, New York, 3. Charles R. Goeldner, J.R. Brent Ritchie (2009): Tourism: Principles, Practices, Philosophies, John Wiley & Sons 4. Christopher M Law (2009): Urban Tourism: The Visitor Economy and the Growth of Large Cities, Continuum Delhi 5. Kamra K.K. and Mohinder Chand (2007): Basics of Tourism: Theory, Operation and Practice, 6. Kanishka Publishers New Delhi 7. Ministry of Tourism (2011): Strategic Action Plan for Tourism in India, Ministry of Tourism, Government of India 8. Prasad, Sheela and Kapil Kumar Gavsner (2016): 'The New 'Love' Story of the Taj Mahal- Urban Planning in the Age of Heritage Tourism in Agra', Economic and Political Weekly, LI(5): 40-48 9. Robinson, H.A. (1996): Geography of Tourism, Macdonald and Evans, London 10. Williams, Stephen (1998): Tourism Geography, Routledge, London
Learning Outcomes	- Students would get the knowledge of concepts of tourism and its typology. - Students will be aware of tourism as an industry and its spatial patterns. - Students would learn the models of ecotourism and the impacts of tourism. - Students would understand the heritage: conservation and revitalization.

Course	B.Sc./B.A.SecondYear:Semester-IV			
CourseTitle	GGRMJ41:Evolution of Geographical Thought			
Category of Course	Major			
Credits&Hourof Teaching		Theory	Practical	Cumulative
	Credits	4		
	Hour of Teaching (Total)	60		
CourseObjectives	- To introduce the students on history of geographical knowledge. - To explain the students about contribution of various scholars and geographers. - To introduce the students about approaches in geography. - To acquaint the knowledge of Indian scholars in geographical knowledge.			
CourseContent	The course content is divided into 4 Units.			
Units	CourseContent			Hr.of Teaching
I	Geography and Geographical Concepts: Early origins of geographical thinking regarding philosophies; Age of exploration and impact of discoveries.			
II	Schools of Geography: Evolution of geographical thinking and disciplinary trends in Germany, France, Britain and United States of America;			
III	Contemporary Themes: Positivism, Quantitative Revolution, Behaviouralism, Humanism, Radicalism, Feminism, Post-Modernism and Post Colonialism.			
IV	Geography in India: Geographical knowledge in ancient India, Teaching and Research in geography during colonial and post-colonial periods; Geography teaching and research in 21 st century: challenges and opportunities; Geography in planning, policy and regional development.			

SEC-Skill Enhancement Course, AEC Ability Enhancement Course; VAC Value Added Course: MD-Multidisciplinary Course 1 Credit (Theory)=15 Hours; 1 Credit (Practical)=30 Hours

Texts/References	1. Daniels, P., Bradshaw, M., Shaw, D. and Sidaway, J. (2000): An Introduction to Human Geography. Issues for the 21st Century. Prentice Hall, London. 2. Dear, M.J. and Flusty, S. (2002): The Spaces of Postmodernity: Readings in Human Geography. Blackwell Publishers, Oxford. 3. Dikshit, R.D. (2004): Geographical Thought. A Critical History of Ideas. Prentice-Hall India, New Delhi. 4. Harvey, M.E. and Holly, P.B. (2002): Themes in Geographic Thought. Rawat Publications, New Delhi. 5. Hubbard, P., Kitchin, R., Bartley, B. and Fuller, D. (2002): Thinking Geographically: Space, Theory and Contemporary Human Geography. Continuum, London. 6. Johnston, R.J. and Sidaway, J.D. (2004): Geography and Geographers. 6th edition, Edward Arnold, London. 7. Martin, G., (2005): All Possible Worlds. A History of Geographical Ideas. 4th edition, Oxford University Press, New York. 8. Peet, R. (1998): Modern Geographical Thought. Blackwell Publishers Inc., Massachusetts. 9. Taylor, G. (ed.) (1953): Geography in the Twentieth Century. Methuen and Company Ltd. and Company, London. 10. Tuan, Yi-Fu (1977): Space and Place. The Perspective of Experience. Edward Arnold, London.
Learning Outcomes	- Students would get the knowledge of geographical ideas. - Students would be aware of various schools of thought. - Students would be able to understand contemporary themes and approaches. - Students would understand the Indian policy and regional development.

Course	B.Sc./B.A.SecondYear:Semester-IV			
CourseTitle	GGRMJ42:Environmental Geography			
Category of Course	Major			
Credits&Hour of Teaching		Theory	Practical	Cumulative
	Credits	4		
	Hour of Teaching (Total)	60		
CourseObjectives	- To introduce the students on nature and scope of environmental geography - To explain the students about man and environment relationships. - To introduce the students to factors of environmental degradation. - To acquaint the knowledge of environmental conservation and management..			
CourseContent	The course content is divided into 4 Units.			
Units	CourseContent			Hr. of Teaching
I	Concept: Nature and scope of environmental geography; Environment and its components; Ecosystem; Biodiversity; Biogeochemical cycles; Ecological succession; Deep Ecology.			15
II	Man-Environment Relationship: Approaches (deterministic, possibilistic and ecological); Historical evolution; Human ecological Adaption; Forms and functions of ecosystems (terrestrial and aquatic).			15
III	Global Environmental challenges: Environmental degradation; pollution; hazards and disaster (Flood Technological); Global and Regional Ecological Change; Urban heat island; Climatic change.			15
IV	Environmental Management: Environmental			15

SEC-Skill Enhancement Course, AECAbility Enhancement Course; VAC Value Added Course: MD-Multidisciplinary Course 1 Credit (Theory)=15 Hours; 1 Credit (Practical)=30 Hours

Texts/References	- Chandna R.C.(2002): Environmental Geography, Kalyani, Ludhiana. - Cunningham W. P. and Cunningham M. A. (2004): Principles of Environmental Science: Inquiry and Applications, Tata Macgraw Hill, New Delhi. - De Blij, H.J.(1996): Human Geography: Culture, Society and Space, 2nd edition. John Wiley and Sons, New York. - Goudie A.(2001): The Nature of the Environment, Blackwell, Oxford. - Huggett, R.J.(1998): Fundamentals of Biogeography, Routledge, London. - Johnston, R.J., Gregory, D., Pratt, G. and Watts, M.(2000): The Dictionary of Human Geography. 4th edition, Basil Blackwell Publishers, Oxford. - Miller G.T.(2004): Environmental Science: Working with the Earth, Thomson Brooks Cole, Singapore. - Odum, E.P. et al.(2005): Fundamentals of Ecology, Cengage Learning India. - Saxena, H.M.(2000): Environmental Management. Rawat Publications., Jaipur and New Delhi. - Singh, S.(2001): Paryavaran Bhugol, Prayag Pustak Bhawan, Allahabad. (in Hindi).
Learning Outcomes	- Students would be able to understand the environmental aspects. - Students would be aware of various types of environments and their functions. - Students would understand the contemporary themes and approaches. - Students would understand environmental management.

Course	B.Sc./B.A.SecondYear:Semester-IV			
CourseTitle	GGRMJ43:Geomorphology			
Category of Course	Major			
Credits&Hourof Teaching		Theory	Practical	Cumulative
	Credits	4		
	Hour of Teaching (Total)	60		
CourseObjectives	• Tointroducethestudentsonnatureandscopeofgeomorphology. • Toexplainthestudentsaboutfundamentalconcepts. • Tointroducethestudentsaboutcontributionsofgeomorphologists. • Toacquainttheknowledgeofregionalgeomorphology...			
CourseContent	Thecoursecontentisdividedinto4Units.			
Units	Course Content			Hr.of Teaching
I	Basic Concepts: Meaning, nature, scope and aspects of geomorphology; Principles and basis of geological time-scale;Fundamental concepts: uniformitarianism and dynamic equilibrium; Relief and differential rate of the geomorphic process.			15
II	Instability of the Earth: Isostasy (Airy and Pratt); plate tectonics;Folded andFaultedstructureand topography, earthquake, volcanicity, masswasting; The cycle of erosion:contributionsofDavisandPenck.			15
III	GradationProcesses: Geomorphicalagentsandprocesses:fluvial,aerial,glacial,peri-glacial, marine and karst; Types and patterns of drainage, Channel: types and pattern.			15
IV	AppliedGeomorphologyandRegionallandscape: Geomorphologyandeconomicdeposits;Geomorphologyinengineeringconstruction;Geomorphologyingroundwaterstudies,Geomorphologyands oil;RegionalgeomorphologyofUttarakhandHimalaya,ChhotanagpurplateauandMiddleGangaPlain			15

SEC-SkillEnhancementCourse, AECAbilityEnhancementCourse; VACValueAdded
Course: MD-MultidisciplinaryCourse1Credit(Theory)=15Hours;1Credit(Practical)=30Hours

Texts/References	1. Bloom,A.L.(1992):Geomorphology–ASystematicAnalysis.Prentice-HallIndia,NewDelhi. 2. Chorley,R.J.,Schumm,S.A.andSugdenD.E.(1984):Geomorphology.Methuen,London 3. Dayal,P.(1994):AtextbookofGeomorphology,KalyaniPublishers,NewDelhi. 4. Holmes,A.(1987):PrinciplesofPhysicalGeology.Nelson,NewYork,3rdedition. 5. Kale,V.andGupta,A.(2001):Elements ofGeomorphology.OxfordUniversityPress,Delhi. 6. Singh,S.(2004): Geomorphology,PrayagPustakBhawan,Allahabad 7. Skinner,B.J.andPorter,S.C.(1996):TheDynamicEarth.JohnWileyandSons,NewYork, 8. Sparks,B.W.(1969):Geomorphology.Longman,London. 9. Stoddard,D.R.(ed.)(1996):ProcessandForminGeomorphology.Routledge,London. 10. Thornbury,W.D.(1990):PrinciplesofGeomorphology.WileyEasternEdition,NewYork.
Learning Outcomes	• Studentswouldbeabletogetknowledgeofbasicconceptsofgeomorphology. • Studentswouldbeawareofthedifferentialrateofthegeomorphic process. • Studentswouldunderstandcontemporarythemesandapproaches. • Studentswouldunderstandtheappliedgeomorphologyandregionallandscape.

Course	B.Sc./B.A.SecondYear:Semester-IV			
CourseTitle	GGRMJ44:Advanced Map Projection and Geological Map			
Category of Course	Major			
Credits&Hourof Teaching		Theory	Practical	Cumulative
	Credits		2	
	Hour of Teaching (Total)		60	
CourseObjectives	- To introduce the students on Classification of map projections. - To explain the students about Elements of geological maps....			
CourseContent				
Units	CourseContent			Hr.of Teaching
I	Map Projection (Mathematical): Meaning, Classification and scope of map projections; Conical: Sinusoidal, Interrupted Sinusoidal and Mollweide; Cylindrical: Mercator's and Gall's projection; International projections			30
II	Geological Map: Elements of geological maps; Beds and bedding plane; Strike, Dip; Completion of outcrop ;geological structure(uniclinal,folded,faulted and unconformable) and their description.			30

SEC-Skill Enhancement Course, AEC Ability Enhancement Course; VAC Value Added Course: MD-Multidisciplinary Course 1 Credit (Theory)=15 Hours; 1 Credit (Practical)=30 Hours

Texts/References	1. Blyth, F.G.H., (1965): Geological Maps and Their Interpretation. Edward Arnold (Publisher) Ltd., London. 2. Hinks, A.R. (1921): Map Projection. Cambridge University Press, London. 3. Mishra, R.N. and Sharma, P.K., (2023): Practical Geography: Methods and Techniques. P areek Publications, Jaipur. 4. Mishra, R.P. and Ramesh A. (1989): Fundamentals of Cartography, Concept Publishing Company, New Delhi. 5. Monkhouse, F.J. and Wilkinson, F.J. (1985): Maps and Diagrams. Methuen, London. 6. Robinson, A.H., Sale, R., Morrison, J. and Muehrcke, P.C. (1984): Elements of Cartography, 5th edition, John Wiley and Sons, New York. 7. Sarkar, A.K. (1997): Practical Geography: A Systematic Approach Orient Longman, Kolkata. 8. Sharma, J.P. (2001): Prayogik Bhugol. Rastogi Publication, Meerut 3rd edition. 9. Singh, R.L. and Singh, Rana P.B. (1993): Elements of Practical Geography (Hindi and English editions). Kalyani Publishers, New Delhi. 10. Steers, J.A. (1965): An Introduction to the Study of Map Projection. University of London Press, London.
Learning Outcomes	- Students would be able to understand the basic concepts of cartography. - Students would be aware of map projection. - Students would be able to understand the geological structure and their description.

Course	B.Sc./B.A.ThirdYear:Semester-V			
CourseTitle	GGRMJ51:Climatology			
Category of Course	Major			
Credits&Hour of Teaching		Theory	Practical	Cumulative
	Credits	4		
	Hour of Teaching (Total)	60		
CourseObjectives	- To acquaint the students about the nature and scope of climatology. - To make the students understand the concept of Heat budget, laws of horizontal motion and air masses. - To develop an understanding of climate change and its impact among students.			
CourseContent	The course content is divided into 4 Units.			
Units	CourseContent			Hr. of Teaching
I	Basics of Climatology: Meaning and scope; Development and recent trends in climatology; Subdivisions of climatology; Elements of weather and climate; Data sources for climatological studies.			15
II	Functions of Atmosphere: Composition and structure of atmosphere; insolation; Heat budget; Air stability; Laws of horizontal motion; general atmospheric circulation; Air Masses: classification and characteristics			15
III	Weather Phenomena: Clouds and Precipitation: formation and types; Fronts: source regions, types and associated weather; Cyclones: tropical and temperate; Monsoon, jet-stream and their significance with reference to India.			15
IV	Climate and Human Response: Climatic classification: Köppen and Thornthwaite; Climate change: causes, evidence and hypotheses, impacts and mitigation			15

SEC-Skill Enhancement Course, AEC Ability Enhancement Course; VAC Value Added Course: MD-Multidisciplinary Course 1 Credit (Theory)=15 Hours; 1 Credit (Practical)=30 Hours

Texts/References	- Barry, R.G. and Carleton, M. (2001): Synoptic and Dynamic Climatology, Routledge, London. - Chorley, R.J. (2001): Atmosphere, Weather and Climate. Methuen, London. - Critchfield, H.J. (2002): General Climatology. Prentice-Hall of India, New Delhi. - Finch, J.C. and Trewartha, G.T.: Elements of Weather and Climate. Prentice-Hall, London. - Kendrew, W.C. (1998): Climatology. Edward Arnold, London. 5th edition. - Lal, D.S. (1986): Climatology. Chaitanya Publications, Allahabad. - Oliver, J.E. and Hidore, J.J. (2003): Climatology: An Atmospheric Science, Pearson Education Private Ltd, Patparganj, Delhi. - Robinson, P.J. and Henderson, S. (1999): Contemporary Climatology, 2nd edition, Pearson Education Ltd., Harlow, UK - Singh, M.B. (1998): Jalvayu Avam Samudra Vigyan. Tara Book Agency, Varanasi. - Singh, M.B. (1999): Jalvayu Avam Jal Vigyan. Tara Book Agency, Varanasi,
Learning Outcomes	- The students will be able to learn about the structure and composition of atmosphere. - They will be able to distinguish between various types of cyclones, and air masses, associated climate and weather changes. - They will develop an understanding of the concept, causes and impact of climate change.

Course	B.Sc./B.A.ThirdYear:Semester-V			
Course Title	GGRMJ52:World Regional Geography			
Category of Course	Major			
Credits & Hour of Teaching		Theory	Practical	Cumulative
	Credits	4		
	Hour of Teaching (Total)	60		
Course Objectives	To introduce the students about region and regionalization. To make the students understand the distribution and characteristics of agricultural and industrial regions of the world. To acquaint the students about the physical, economic and cultural aspects of Japan, Brazil and select countries of South Asia.			
Course Content	The course content is divided into 4 Units.			
Units	Course Content			Hr. of Teaching
I	Concepts: Region and Regionalization; Bases of Regionalization; Types of Regions; Bases & Approaches of classification and characteristics of Developed and Developing World; Levels of Development: Developed, Developing, Under-developed and Least developed World			15
II	Major Regions of the World: Natural regions, Climatic regions, Cultural regions, Agricultural regions, Industrial regions, Population resource regions			15
III	Regional Study of Japan and Brazil: Physiography, Climate, Soil, Vegetation, Mineral, Power Resources (Iron ore and Coal), Industries and Population			15
IV	Regional study of South Asia: Physiography, Climate, Soil, Vegetation, Mineral and Power Resources, Population, Strategic Importance of South Asia.			15

SEC-Skill Enhancement Course, AEC Ability Enhancement Course; VAC Value Added Course; MD-Multidisciplinary Course
1 Credit (Theory) = 15 Hours; 1 Credit (Practical) = 30 Hours

Texts/References	1. Blij, H. J. De, (1971): Geography: Regions and Concepts, John Wiley and Sons. 2. Claval P. I., (1998): An Introduction to Regional Geography, Black well Publishers, Oxford and Massachusetts. 3. Friedmann J. and Alonso W., (1975): Regional Policy-Readings in Theory and Applications, MIT Press, Massachusetts. 4. Gavsker. K. K. (2014). Conceptualizing the <i>Region</i>: An Overview of Recent Developments and Relevance, International Journal of Innovative Research and Practices, 2(12):8-20 5. Gore C. G., (1984): Regions in Question: Space, Development Theory and Regional Policy, Methuen, London. 6. Gore C. G., Köhler G., Reich U. P. and Ziesemer T., (1996): Questioning Development; Essays on the Theory, Policies and Practice of Development Intervention, Metropolis- Verlag, Marburg. 7. Haynes J., (2008): Development Studies, Polity Short Introduction Series. 8. Johnson E. A. J., (1970): The Organization of Space in Developing Countries, MIT Press, Massachusetts. 9. Peet R., (1999): Theories of Development, The Guilford Press, New York 10. Rai, G., (2016): Regional Geography of the World (<i>in Hindi</i>), Wisdom Books, Varanasi 11. Rai, G., (2019): Asia: Regional Geography (<i>in Hindi</i>), Wisdom Publications, Varanasi. 12. Rai, G., (2019): Regional Study of South East Asia (<i>in Hindi</i>), Wisdom Publications, Varanasi.
Learning Outcomes	• The students will learn about the basis of classification and characteristics of a region. • They will get familiarized with the distinguishing characteristics of developed, developing and under-developed countries. • They will learn the concept and characteristics of population resource region. • They will gain knowledge of the physical, economic and cultural aspects of Japan, Brazil and select countries of South Asia.

Course	B.Sc./B.A.ThirdYear:Semester-V			
CourseTitle	GGRMJ53:Population Geography			
Category of Course	Major			
Credits&Hourof Teaching		Theory	Practical	Cumulative
	Credits	4		
	Hour of Teaching (Total)	60		
CourseObjectives	• To make the students understand meaning, scope and development of population geography. • To familiarize them with the concept, determinants and characteristics of fertility, mortality and migration. • To make them understand the population characteristics and policies of India. • To familiarize them with the trends and distribution of world population, urbanization and population problems.			
CourseContent	The course content is divided into 4 Units.			
Units	CourseContent			Hr.of Teaching
I	Basic Concepts: Meaning, scope and development of population geography; Relationship of Population geography with other branches of social sciences; Sources and types of population data: census, sample survey, vital registration system.			15
II	Population Dynamics: Measures, trends and pattern of Fertility and Mortality; Concept, types, determinants and consequences of Migration; Theories of population growth: Malthusian, Demographic Transition Theory;			15
III	World Population: Population distribution and growth; Population agglomerations; Urbanization trends and patterns; Human Resource Development: indicators and patterns; Population problems.			15
IV	Population Characteristics in India: Population distribution, Density And Growth; Literacy; Age and Sex composition; Population Policies of India.			15

SEC-Skill Enhancement Course, AEC Ability Enhancement Course; VAC Value Added Course: MD-Multidisciplinary Course 1 Credit (Theory)=15 Hours; 1 Credit (Practical)=30 Hours

Texts/References	1. Chandana, R.C. (2006): Geography of population. Kalyani Publisher New Delhi. 2. Clarke, J.I. (1972): Population Geography. Pergamon Press, Oxford. 3. Demko, G.J., Rose, H.M., and Schnell, G.A. (1970): Population Geography. A Reader MNG raw-Hill, New York. 4. Dube, K.K. and Singh, M.B. (1994): Jansankhya Bhoogol, Rawat Publications, Jaipur. 5. Garnier, B.J. (1993): Geography of Population. 3rd edition, Longman, London. 6. Jones, H.R. (1981): A Population Geography. Harper and Row, New York. 7. Peters, G.L. and Larkin, R.P. (1983): Population Geography: Problems, Concepts and Prospects. Kendall/Hunt, Dubuque, IA. 8. Trewartha, G.T. (1985): A Geography of Population: World Patterns. John Wiley and Sons, New York. 9. Zelinsky, W. (1966): A Prologue to Population Geography, Prentice Hall, New Jersey.
Learning Outcomes	• Students would be able to understand the development of population geography • Students would be able to learn about Population agglomerations. • They will be Acquainted the knowledge of Population problems. • Students would be able to understand the regional variations.

Course	B.Sc./B.A.ThirdYear:Semester-V			
CourseTitle	GGRMJ54:CartographicTechniques			
Category of Course	Major			
Credits&Hourof Teaching		Theory	Practical	Cumulative
	Credits		2	
	Hour of Teaching (Total)		60	
CourseObjectives	- To introduce the students on Nature and types of data. - To explain the students about Distribution maps.			
CourseContent	The course content is divided into 2 Units.			
Units	CourseContent			Hr.of Teaching
I	Representation of Data: Nature and types of data; Representation of data: bar diagram, histogram, frequency curve, line graph, ogive, pie chart; Traffic flow diagram; Slope analysis: Smith and Wentworth methods.			30
II	Distribution Maps and Diagrams: Rain fall dispersion diagram, water balance graph Distribution maps: dot, isopleth, choropleth, chorochromatic and chorochromatic. Graded and nongraded shadings.			30

SEC-Skill Enhancement Course, AEC Ability Enhancement Course; VAC Value Added Course; MD-Multidisciplinary Course
1 Credit (Theory)=15 Hours; 1 Credit (Practical)=30 Hours

Texts/References	1. Blyth, F.G.H., (1965): Geological Maps and Their Interpretation. Edward Arnold (Publisher) Ltd., London. 2. Hinks, A.R. (1921): Map Projection, Cambridge University Press, London. 3. Mishra, R.N. and Sharma, P.K., (2023): Practical Geography: Methods and Techniques. P areek Publications, Jaipur. 4. Mishra, R.P. and Ramesh A. (1989): Fundamentals of Cartography, Concept Publishing Company, New Delhi. 5. Monkhouse, F.J. and Wilkinson, F.J. (1985): Maps and Diagrams. Methuen, London. 6. Robinson, A.H., Sale, R., Morrison, J. and Muehrcke, P.C. (1984): Elements of Cartography. 5th edition, John Wiley and Sons, New York. 7. Sarkar, A.K. (1997): Practical Geography: A Systematic Approach Orient Longman, Kolkata. 8. Sharma, J.P. (2001): Prayogik Bhugol. Rastogi Publication, Meerut 3rd edition. 9. Singh, R.L. and Singh, Rana P.B. (1993): Elements of Practical Geography. (Hindi and English editions). Kalyani Publishers, New Delhi. 10. Steers, J.A. (1965): An Introduction to the Study of Map Projection. University of London Press, London.
Learning Outcomes	- Students would be able to understand the knowledge on basic concepts of cartography. - They will be able to know about one and two standard parallels. - Students would be able to understand the geological structure and their description.

Course	B.Sc./B.A.ThirdYear:Semester-V			
CourseTitle	GGRMV51:Hazards & Disaster Risk Reduction			
Category of Course	Minor Voc			
Credits&Hour of Teaching		Theory	Practical	Cumulative
	Credits	4		
	Hour of Teaching (Total)	60		
CourseObjectives	- To introduce the meaning, concepts, and definitions of disaster. - To explain disaster management cycle: prevention, mitigation and preparedness. - To introduce the students about environmental health. - To acquaint the knowledge of risk management in India.			
CourseContent	The course content is divided into 4 Units.			
Units	CourseContent			Hr.of Teaching
I	Introduction to Disasters: Concepts, and definitions (Disaster, Hazard, Vulnerability, Resilience, Risks; Disasters: Classification; Causes, Impacts & Approaches to Disaster Risk Reduction: Social, economic, political, environmental, health, psychosocial, etc.			15
II	Differential impacts: in terms of caste, class, gender, age, location, disability Global trends in disasters; Disaster management cycle: prevention, mitigation and preparedness; myths and social responses.			15
III	Disaster Risk Management in India: Community based Disaster Risk Reduction, Structural- nonstructural measures; Hazard and vulnerability profile of India, Institutional arrangements, Disaster mitigation.			15
IV	Case Studies: Landslide, Cloud burst, Mitigation, Response and Preparedness, Disaster Mitigation Act and Policy, Other related policies, plans, programs and legislation.			15

SEC-Skill Enhancement Course, AEC Ability Enhancement Course; VAC Value Added

Course: MD-Multidisciplinary Course 1 Credit (Theory)=15 Hours; 1 Credit (Practical)=30 Hours

Texts/References	- Alexander David, Introduction in 'Confronting Catastrophe', Oxford University Press, 2000 - Andharia J. Vulnerability in Disaster Discourse, JTCDM, Tata Institute of Social Sciences Working Paper no. 8, 2008 - Blaikie, P, Cannon T, Davis I, Wisner B 1997. At Risk Natural Hazards, Peoples' Vulnerability and Disasters, Routledge. - Coppola P, Damon, 2007. Introduction to International Disaster Management, - Carter, Nick 1991. Disaster Management: A Disaster Manager's Handbook. Asian Development Bank, Manila Philippines. - Cuny, F. 1983. Development and Disasters, Oxford University Press. Document on World Summit on Sustainable Development 2002. - Govt. of India: Disaster Management Act 2005, Government of India, New Delhi. Government of India, 2009. National Disaster Management Policy, - Gupta Anil K, Sreeja S. Nair. 2011 Environmental Knowledge for Disaster Risk Management, NIDM, New Delhi - Kapur, Anu & others, 2005: Disasters in India Studies of grim reality, Rawat Publishers, Jaipur - Kapur Anu 2010: Vulnerable India: A Geographical Study of Disasters, IIA Sand Sage Publishers, New Delhi.
Learning Outcomes	- Students would be able to understand the knowledge of Disaster, Hazard, Vulnerability. - Students would be able to understand the knowledge about Social, economic, political environment. - They will be Acquainted the knowledge of Population problems with disaster. - Student should understand the Hazard and Vulnerability profile of India.

Course	B.Sc./B.A.ThirdYear:Semester-V			
CourseTitle	GGRIN51: Internship			
Category of Course	Internship-1			
Credits&Hourof Teaching		Theory	Practical	Cumulative
	Credits	2		
	Hour of Teaching (Total)	30		

Course	B.Sc./B.A.ThirdYear:Semester-VI			
CourseTitle	GGRMJ61: Advanced Remote Sensing and Geographical Information System			
Category of Course	Major			
Credits&Hourof Teaching		Theory	Practical	Cumulative
	Credits	4		
	Hour of Teaching (Total)	60		
CourseObjectives	- To introduce the Meaning, Concepts, and basics of remote sensing. - To explain Characteristics and spectral bands. - To introduce the students about remote sensing processes. - To acquaint the knowledge of satellite data interpretation.			
CourseContent	The course content is divided into 4 Units.			
Units	CourseContent			Hr. of Teaching
I	Basics of Remote Sensing: Definition, scope, and history of Development; Types of Remote Sensing: basis and classification; Electromagnetic Radiation (EMR): Characteristics and spectral bands; EMR interaction with the Atmosphere, and Earth's Surface (Land, soil and water); Atmospheric Windows; Spectral Signature; Acquisition of remote sensing data			15
II	Remote Sensing Processes: Platforms and Sensors: Types and characteristics; Aerial Photographs: Types and its geometry; Remote Sensing Data: Types, characteristics and sources; Image Resolution; Satellite Data Interpretation: Visual and digital image interpretation; Interpretation keys, Techniques in image interpretation			15
III	Fundamentals of GIS: Definition, Evolution and Components of Geographical information system; Recent advances in Geographical Information System; Coordinate systems, Datums and projections in Geographical Information System; Raster and Vector data models; GIS data quality, errors, input, geo-referencing, digitization, editing, query, and output. Types and characteristics of map, contents and layout in Geographical Information System.			15
IV	Applications of Remote Sensing and Geographical Information System: Land use/Land Cover Mapping, Geological and geomorphological mapping and water resource management.			15

SEC-Skill Enhancement Course, AEC Ability Enhancement Course; VAC Value Added Course: MD-Multidisciplinary Course 1 Credit (Theory)=15 Hours; 1 Credit (Practical)=30 Hours

Texts/References	- Bolstad, P. (2005). GIS Fundamentals: A First Text on Geographic Information Systems. Elsevier Press. - Elangovan, K. (2006). GIS: Fundamentals, Applications and Implementations. New India Publishing. - Jensen, J. R. (2009). Remote Sensing of The Environment: An Earth Resource Perspective 2/E. Pearson Education India. - Lillesand, T., Kiefer, R. W., & Chipman, J. (2014). Remote Sensing and Image Interpretation. John Wiley & Sons.
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	5. Panigrahi, N. (2009). Geographical Information Science. Universities Press. 6. Rees, W. G. (2012). Physical Principles of Remote Sensing. Cambridge University Press. 7. Sabins, F. F. (2007). Remote Sensing: Principles and Applications. Waveland Press.
Learning Outcomes	- Students would be able to learn the knowledge of response pattern of earth features. - Students would be able to aware about platforms and sensors. - They will be acquainted about visual and digital image interpretation. - Students should understand the spatial and non-spatial data attributes.

Course	B.Sc./B.A. Third Year: Semester-VI			
Course Title	GGRMJ62: Oceanography			
Category of Course	Major			
Credits & Hour of Teaching		Theory	Practical	Cumulative
	Credits	4		
	Hour of Teaching (Total)	60		
Course Objectives	- To introduce the Meaning, Concepts, and basics of remote sensing. - To explain Characteristics and spectral bands. - To introduce the students about remote sensing processes. - To acquaint the knowledge of satellite data interpretation.			
Course Content	The course content is divided into 4 Units.			
Units	Course Content			Hr. of Teaching
I	Basic Concepts: Meaning and scope of oceanography; Origin of oceans; Ocean basins: continental shelf, continental slope, deep sea plains, ocean deeps, submarine canyons.			15
II	Properties and Circulation: Physical and chemical properties of sea water: temperature, density, salinity; Circulation of ocean water: tides and currents.			15
III	Marine Life and Deposits: Classification and distribution of marine deposits, Coral reefs: origin, types, characteristics and theories.			15
IV	Human Ocean Relation: Mineral and energy resources; Indian Ocean Realm; Geo-politics of Indian ocean; Maritime security; Global warming and sea level change.			15

SEC-Skill Enhancement Course, AEC Ability Enhancement Course; VAC Value Added Course; MD-Multidisciplinary Course 1 Credit (Theory)=15 Hours; 1 Credit (Practical)=30 Hours

Texts/References	- Cuchlaine A. M. King, (1965), Oceanography for geographers, Edward Arnold Ltd, London - Gerhard Neumann, Willard J. Pierson, Jr. (1966), Principles Physical Oceanography, Prentice-Hall, Inc. Englewood Cliffs, N.J. - Lal, D. S. (2009), Climatology and Oceanography, Sharda Pustak Bhavan, Allahabad. - MN Pherson, Kenneth. (1993). The Indian Ocean: A History of People and the Sea. Oxford: Oxford University Press - Pearson, Michael. (2003). The Indian Ocean. Routledge, London. - Sverdrup, Harald Ulrik; Johnson, Martin Wiggo; Fleming, Richard H. (1942). The Oceans, Their Physics, Chemistry, and General Biology. New York: Prentice-Hall. - Sverdrup, Keith A., Duxbury, Alyn C., Duxbury, Alison B. (2006). Fundamentals of Oceanography, MNGraw-Hill - Talley L. D., Pickard G. L., Emery W. J. and Swift J. H. (2011). Descriptive Physical Oceanography: An Introduction (Sixth Edition Elsevier, Boston. - Vatal, M., Sharma, R. C. (2011), Chaitanya Publishing House, Allahabad.
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Learning Outcomes	• Students would be able to understand the knowledge of submarine canyons. • Students would be able to learn the knowledge about Circulation of ocean water. • They will be acquainted about the knowledge of Marine Life and Deposits. • Students would be able to understand to Global warming and sea level change.
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Course	B.Sc./B.A.ThirdYear:Semester-VI			
Course Title	GGRMJ63: Remote Sensing and Geographical Information System Techniques			
Category of Course	Major			
Credits & Hour of Teaching		Theory	Practical	Cumulative
	Credits		2	
	Hour of Teaching (Total)		60	
Course Objectives	• To introduce the Meaning, Concepts, and basics of remote sensing. • To explain Characteristics and spectral bands. • To introduce the students about remote sensing processes. • To acquaint the knowledge of satellite data interpretation.			
Course Content	The course content is divided into 2 Units.			
Units	Course Content			Hr. of Teaching
I	Remote Sensing Data Collection and Interpretation: Data sources; Elements of Aerial Photo and Satellite Image Interpretation; Preparation of the maps viz. Land use/ Land cover, and geomorphological mapping			30
II	Geographical Information System Data and its Analysis: Data sources: Bhuvan; Image Registration, Creation of Personal Geodatabase: Digitization and creation of shape files: Point, Line and Polygon, adding attributes Data error corrections; Spatial and non-Spatial Data Query; Spatial analysis methods (overlay, buffering, and interpolation)			30

SEC-Skill Enhancement Course, AECAbility Enhancement Course; VAC Value Added Course: MD-Multidisciplinary Course 1 Credit (Theory)=15 Hours; 1 Credit (Practical)=30 Hours

Texts/References	1. Claudia, K., Stefan, D., 2014. Quantitative Remote Sensing in Thermal Infrared, 11, Springer, 281p 2. De- Jong, Steven, M Vander Meer, F.D., 2004. Remote Sensing Image Analysis: Including the Spatial Domain: Including the Spatial Domain, 5, Springer, 359p. 3. James B. Campbell & Randolph H. Wynne., Introduction to Remote Sensing, The Guilford Press, 2011. 4. Rencz, A.N., 2008. Remote Sensing for the Earth Sciences, Manual of Remote Sensing, 3, A SPRS, 703p. Van-dr-Meer, F., De Jong, S., 2006. Imaging spectrometry: Basic principles and prospective applications (The Netherlands: Springer Publishers), 451p.
Learning Outcomes	• Students would be able to understand the knowledge of pattern of earth features. • Students would be able to learn the knowledge about platforms and sensors • Acquainted the knowledge of visual and digital image interpretation. Students should understand the Spatial analysis methods.

Course	B.Sc./B.A.ThirdYear:Semester-VI			
Course Title	GGRMJ64:Quantitative Methods in Geography			
Category of Course	Major			
Credits&Hour of Teaching		Theory	Practical	Cumulative
	Credits		2	
	Hour of Teaching (Total)		60	
CourseObjectives	• To introduce the frequency distribution. • To explain measures of dispersion. • To introduce the students about correlation.			
CourseContent	The course content is divided into 2 Units.			
Units	CourseContent			Hr.of Teaching
I	Central Tendency and Dispersion: Frequency distribution; Mean, median, mode; Measures of dispersion: mean deviation, standard deviation, Coefficient of variation.			30
II	Correlation and Regression: Correlation (Karl Pearson and Spearman) and Scatter diagram; Regression analysis; Location quotient, Lorenz curve, Gini Coefficient.			30

SEC-Skill Enhancement Course, AECAbility Enhancement Course; VAC Value Added Course: MD-Multidisciplinary Course 1 Credit(Theory)=15Hours; 1 Credit(Practical)=30Hours

Texts/References	1. Bhagwathi, V. and Pillai, R. S. N. (2003): Practical Statistics, Sultan Chand and Company, New Delhi 2. Ebdon, D. (1977): Statistics in Geography: A Practical Approach, Blackwell Publishers Inc., Massachusetts 3. Gregory, S. (1973): Statistical Methods and the Geographer, Longman, London. 4. Gupta, S. P. (1998): Advanced Practical Statistics, Sultan Chand and Company, New Delhi 5. Mahmood, A. (1986): Statistical Methods in Geographical Studies, Rajesh Publications, New Delhi 6. Zamir, A. (2002): Statistical Geography: Methods and Applications, Rawat Publications, Jaipur.
Learning Outcomes	• Students would be able to understand the knowledge of mean deviation, standard deviation. • They will be made aware about mean, median, mode. • They will be acquainted for the knowledge of Location quotient

Course	B.Sc./B.A.ThirdYear:Semester-VI			
Course Title	GGRMJ65A:Hydrology			
Category of Course	Major			
Credits & Hour of Teaching		Theory	Practical	Cumulative
	Credits	4		
	Hour of Teaching (Total)	60		
Course Objectives	- To introduce the meaning and scope of Hydrology. - To explain the process of evaporation. - To introduce the students about factors controlling infiltration. - To acquaint the knowledge of Stream flow measurement.			
Course Content	The course content is divided into 4 Units.			
Units	Course Content			Hr.of Teaching
I	Bases of Hydrology: Nature, scope and development to Hydrology; Hydrological cycle; Man's Influence on the hydrological cycle; Precipitation: formation process, types, characteristics and measurements.			15
II	Water Losses: Interception: factors Affecting			15
III	Water Disposition: Infiltration: infiltration process and factors controlling infiltration; Soil moisture and its zones: storage of soil moisture, factors affecting soil moisture storage; Runoff: its sources and components, factors affecting runoff; River regimes.			15
IV	Assessment and Conservation: Stream flow measurement: interpretation of stream flow data, Hydrograph: components and separation; Ground water: occurrence, storage, recharge and discharge; Floods: estimation and control; Appraisal of water resources; Water policies and conservation in India.			15

SEC-Skill Enhancement Course, AEC Ability Enhancement Course; VAC Value Added Course; MD-Multidisciplinary Course 1 Credit (Theory)=15 Hours; 1 Credit (Practical)=30 Hours

Texts/References	1. Chow, V.T., D.R. Maidment and L.W. Mays (1988): Applied Hydrology. MNGraw-Hill 2. Dingman L.S. (2002): Physical Hydrology. 2nd ed. Waveland Press Inc., USA. 3. Rai, V.K. (1993): Water Resource Planning and Development, Deep and Deep Publication, New Delhi 4. Hann C.T. (1995): Statistical Methods in Hydrology. First East-West Press Edition, New Delhi. 5. Singh, M.B. (1999): Climatology and Hydrology. Tara Book Agency, Varanasi. (In Hindi). 6. Ward, R.C. and Robinson, M. (2000): Principles of Hydrology. MNGraw Hill, New York. 7. Subramanya K. (2004): Engineering Hydrology. Tata MNGraw-Hill, New Delhi. 8. Jain S.K., Agarwal P.K. and Singh V.P. (2007): Hydrology and Water Resources of India. Springer, The Netherlands. 9. Raghunath H.M. (2006): Hydrology. New Age International (P) Ltd., New Delhi. 10. Shaw E.M. (2004): Hydrology in Practice. 3rd ed, Routledge Publishers, London. 11. Singh G., Venkataraman C., Sastry G. and Joshi B.P. (1990): Manual of Soil and Water Conservation Practices. Oxford and
Learning Outcomes	- Students would be able to understand the knowledge of submarine canyons. - Students would be able to learn the knowledge about Circulation of ocean water. - Students would be able to understand the knowledge of Marine Life and Deposits. - Students should understand the Global warming and sea level change.

Course	B.Sc./B.A.ThirdYear:Semester-VI			
CourseTitle	GGRMJ65B:Geography of Settlement			
Category of Course	Major			
Credits&Hourof Teaching		Theory	Practical	Cumulative
	Credits	4		
	Hour of Teaching (Total)	60		
CourseObjectives	- To introducethe meaning and scope of settlement geography. - To explain the processes of settlement growth. - To introduce the students about factors affecting settlement. - To acquaint the knowledge of Concept of urban planning.			
CourseContent	The course content is divided into 4 Units.			
Units	CourseContent			Hr.of Teaching
I	Conceptual Introduction: Defining human settlements; Nature and scope of Settlement Geography; Concept of rural and urban Settlements; Origin and Growth of urban settlements; Concept of megacities, global city and edge city.			15
II	Location, Morphology and Growth: Site, Situation, Types and Patterns; Internal Morphology of Rural and Urban Settlements; Ecological Processes of Settlement Growth; Rural-urban fringe; Umland and hinterland			15
III	Distributional Characteristics of Settlements: Size and spacing of cities: Rank-size rule, Law of Primate City; Functional classification of urban settlements; Settlement Hierarchy-Christaller's Central Place Theory; August Losch's Theory			15
IV	Urban Region and Planning Characteristics of Metropolitan region - A case of Delhi and Varanasi; Rural Housing Planning and Policies; Concept of urban planning			15

SEC-Skill Enhancement Course, AEC Ability Enhancement Course; VAC Value Added Course: MD-Multidisciplinary Course 1 Credit (Theory)=15 Hours; 1 Credit (Practical)=30 Hours

Texts/References	- Chisholm, M. (1970) Rural Settlement and Land Use. London: Hutchinson. - Daniel, P. (2002). Geography of Settlement, New Delhi: Rawat Publications, Jaipur. - Doxiadis, C. A. (1968). An Introduction to Science of Human Settlements, - Hudson, F. S. (1976): A Geography of Settlements. Mac Donald and Evans, New York. - Money, D. C. (1972). Patterns of Settlements, London: Evan Brothers - Ramachandran, R., (1991): Urbanisation and Urban Systems in India. Oxford University Press, Delhi. - Singh R. Y., 1994: The Geography of Settlement, Rawat Publication, New Delhi. - Stanley, B., Jack, W. and Donald, Z. (eds.) (2003): Cities of the World. Rowman and Littlefield, New York and Oxford. - Wood, M. (2005): Rural Geography: Processes, Responses and Experiences of Rural Restructuring. Sage Publication, London.
Learning Outcomes	- Students would be able to understand the knowledge of settlement and its patterns. - Students would be able to learn about Rural and Urban Settlements. - Acquainted the knowledge of Functional classification of urban settlements. - Students should understand the Characteristics of Metropolitan region.

Course	B.Sc./B.A.ThirdYear:Semester-VI			
CourseTitle	GGRMJ65C:Resource Geography			
Category of Course	Major			
Credits&Hour of Teaching		Theory	Practical	Cumulative
	Credits	4		
	Hour of Teaching (Total)	60		
CourseObjectives	• To introduce the meaning and scope of resource geography. • To explain the distribution and pattern of resources • To introduce the natural resources in India. • To acquaint the theories of agricultural and industrial location.			
CourseContent	The course content is divided into 4 Units.			
Units	Course Content			Hr. of Teaching
I	Basic Concepts: Nature and scope of Resource Geography; Concept and classification of resources; Distribution and pattern of resources: Land, water, minerals (Iron ore, Bauxite, Copper) and power resources; Non-conventional sources of energy, Scientific and technological innovation and resources.			15
II	Human Resource and Locational Aspects: Human Resource: concept, classification, determinants and problems; Theories of agricultural and industrial location: Von Thunen, Weber, Losch and Hoover			15
III	Problems and Conservation of Resource: Nature and Production of Resources; Resource Scarcity; World energy utilization and its crisis; Sustainable development, Resource conservation; National Policies and Programmes on Natural Resources in India: Land, Forest and Water			15
IV	Regional Perspectives: World economic development; Concept of developed and developing nations; Concepts of North-South and First, Second, Third and Fourth Worlds			15

SEC-Skill Enhancement Course, AEC Ability Enhancement Course; VAC Value Added Course: MD-Multidisciplinary Course 1 Credit (Theory)=15 Hours; 1 Credit (Practical)=30 Hours

Texts/References	1. Burton, I. and Kates, R. W. (1978): Readings in Resource Management and Conservation. MNGraw Hills, New York 2. Clark, G. L., Feldman, M. P. and Gertler, M. S. (eds.) (2000): The Oxford Handbook of Economic Geography. Oxford University Press, Oxford and New York. 3. Ehrlich, P. R., Ehrlich, R. H. and Holdren, J. P. (1998): Ecoscience: Population, Resources and Development. 2nd edition. Freeman and Company, San Francisco. 4. Sheppard E. and Treror I. B. (ed.) (2003): A Companion to Economic Geography, Blackwell Publication, U.K. and USA. 5. MM Jarty, H. M. and James B. L. (1976): A Preface to Economic Geography. Prentice Hall, New Jersey. 6. Mitra, A. (2000): Resource Studies; Shridhar Publishers., Kolkata. 7. Ramesh, A. (ed.) (1984): Resource Geography. Heritage Publishers, New Delhi. 8. Singh, J. (2000): Sansadhaan Bhoogol, Gyanodaya Prakashan, Gorakhpur 9. Singh, K. N. and Singh, J. (2003): Arthik Bhoogol Ke Mool Tatva, Gyanodaya Prakashan, Gorakhpur. 10. Todaro M. P. and Smith S. C. (2004): Economic Development, Pearson Education, (Singapore) Private Ltd. Singapore
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Learning Outcomes	- Students would be able to understand the knowledge of settlement and its patterns. - Students will be made aware about rural and urban settlements. - Acquainted the knowledge of Functional classification of urban settlements. - Student will understand the characteristics of metropolitan region.
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Course	B.Sc./B.A.Third Year: Semester-VI			
Course Title	GGRMJ65D: Social Geography			
Category of Course	Major			
Credits & Hour of Teaching		Theory	Practical	Cumulative
	Credits	4		
	Hour of Teaching (Total)	60		
Course Objectives	- To introduce the meaning and scope of social geography. - To explain the social differentiation and spatial justice - To introduce the language and religion in India. - To acquaint the tribes and their evolution.			
Course Content	The course content is divided into 4 Units.			
Units	Course Content			Hr. of Teaching
I	Concept and Approaches: Meaning of social geography; Concepts: society, community, social structure and relations; Social differentiation and stratification: space, social space, social morphology, social-spatial, spatial justice, Spatial inclusion and exclusion;			15
II	Social differentiation and region formation: Bases of social region formation: ethnicity, tribes, dialect, language and religion.			15
III	Regional basis of Indian Society: Peopling of India; Evolution of social regions; Tribes: Characteristics and their distribution; Caste: concept, characteristics and classification, Linguistic regions of India; Particularly Vulnerable Tribes Groups (PVTGs), Unity in diversity.			15
IV	Geography of Social Well-being in India: Concepts of social wellbeing; Social geographical perspectives on crime, healthcare, education and housing; Geography of men and women; Social impact assessment; Social policies and planning.			15

SEC-Skill Enhancement Course, AEC Ability Enhancement Course; VAC Value Added

Course: MD-Multidisciplinary Course 1 Credit (Theory)=15 Hours; 1 Credit (Practical)=30 Hours

Texts/References	- Ahmad, A. (1999): Social Geography. Rawat Publications, Jaipur and New Delhi. - Anderson, K. (2006): Race and Crises of Human Development. Routledge, London and New Delhi. - Betizlle, A. (1983): Equality and Inequality, Oxford University Press, New Delhi - Brewer, J.D. (2000): Ethnography, Open University Press, Buckingham, Philadelphia, First Edition. - Coates, B.E., Johnston, R.J. and Knox, P.L. (1977): Geography and Inequality, Oxford University Press, Oxford, London. - Eyles, J. (ed.) (1986): Social Geography in International Perspective. Rowman and Littlefield, New Jersey and Los Angeles. - Dubey, S.C. (1991): Indian Society. National Book Trust, New Delhi. - Forde, C.D. (1934): Habitat, Economy and Society, Methuen and Company, London - Gregory, D. and Larry, J. (eds.) (1985): Social Relations and Spatial Structures. MN Millan, London - Gore, M.S. (1985): Social Aspects of Development, Rawat Publications, Jaipur
Learning Outcomes	- Students would be able to understand the knowledge of social groups. - Students would be able to understand about community, social structure and relations. - They will be acquainted the knowledge of Functional classification. - Students should understand the linguistic regions of India.

Course	B.Sc./B.A.ThirdYear:Semester-VI			
Course Title	GGRMJ65E:Cultural Geography			
Category of Course	Major			
Credits&Hour of Teaching		Theory	Practical	Cumulative
	Credits	4		
	Hour of Teaching (Total)	60		
CourseObjectives	• To introduce the meaning and scope of cultural geography. • To explain the culture, society and civilization • To introduce the Cultural turn in human geography. • To acquaint the Geography of food and house type.			
CourseContent	The course content is divided into 4 Units.			
Units	CourseContent			Hr.of Teaching
I	Concepts and Approaches: Meaning of cultural geography; Concepts: culture, society, civilization, cultural hearth, cultural area, core, domain, sphere, cultural realm, place-making and cultural landscape; Approaches to the study of cultural geography;			15
II	Evolution of Culture: Evolution of early human groups; Beginning of civilizations; Process of formation of cultural regions; cultural regions of the world, vernacular regions; Evolution of cultural landscapes: globalization and culture.			15
III	Cultural processes: Language and dialect: Classification and regions, language and identity, language and nationalism; Language in the landscape: Religions: religion, identity and conflict, religious landscapes			15
IV	Geographical basis of Indian Culture: Linguistic and dialect regions of India; Multi lingual states; Folk cultures; Geography of food and house type; Geography pilgrimage; Cultural regions of India.			15

SEC-Skill Enhancement Course, AEC Ability Enhancement Course; VAC Value Added

Course: MD-Multidisciplinary Course 1 Credit (Theory)=15 Hours; 1 Credit (Practical)=30 Hours

Texts/References	1. Appadurai, A. 1996: Modernity At Large: Cultural Dimensions Of Globalization. Minneapolis: University of Minnesota Press. 2. Barber, C.L. 1965. The Story of Language. London: The English Language Book Society and Pan Books Ltd. 3. Bhardwaj, S.M. 1973. Hindu Places of Pilgrimage in India. Berkeley: University of California Press. 4. Blunt, A., Gruffudd, P., May, J. and Ogborn, M. 2003: Cultural geography in practice. London: Arnold. 5. Crang, M. 1998: Cultural Geography. London: Routledge. 6. Kosambi, D.D. 1972. The Culture and Civilization of Ancient India in Historical Outline. New Delhi: Vikas Publications. 7. Maloney, C. 1974. People of South Asia. New York: Winston Norton. 8. Norton, W. 2006. Cultural Geography. Environments. Landscapes. Identities. Inequalities (2nd ed.). Toronto: Oxford University Press. 9. Parekh, B. 2000: Multiculturalism: Cultural Diversity and Political Theory. London: Macmillan Press. 10. Sopher, David E., ed. 1980. An Exploration of India: Geographical Perspectives on Society and Culture. Ithaca, New York: Cornell University Press.
Learning Outcomes	• Students would be able to understand the knowledge of cultural landscape. • Students would be able to understand the knowledge about Evolution of early human groups. • Students would be able to learn language and nationalism. • Students should understand the Linguistic and dialect regions.

Course	B.Sc./B.A.ThirdYear:Semester-VI			
Course Title	GGRMV61:Geography of Health and Well being			
Category of Course	MinorVoc			
Credits & Hour of Teaching		Theory	Practical	Cumulative
	Credits	4		
	Hour of Teaching (Total)	60		
Course Objectives	- To introduce the meaning and scope of health geography. - To explain the determinants of health quality - To introduce the geographical inequality in provision of healthcare, - To acquaint the disease ecology, Impact of clean air, adequate water.			
Course Content	The course content is divided into 4 Units.			
Units	Course Content			Hr. of Teaching
I	Geography of Health as a sub-discipline of Human Geography: Definition, scope, approaches, historical development and key contributors ; Determinants of health; Quality of life: Concept and measurement; Disease burden; Epidemiological transition.			15
II	Spatial Patterns: Disease burden at global, regional and national scales; Globalization, migration, and spatial diffusion of diseases; Geographical inequality in provision, access, and use of health care services.			15
III	Environment and Health: Disease ecology, Impact of clean air, adequate water, improved sanitation and hygiene on health; built environment and health; Therapeutic landscapes; climate change and health.			15
IV	Indian Context: Public health system, National health policy, Major health and disease control programs: National Programme for Prevention & Control of Non-Communicable Diseases (NP-NCD), Revised National Tuberculosis Control Programme (RNTCP), National Vector Borne Disease Control Programme, National Health Mission, Ayushman-Bharat.			15

SEC-Skill Enhancement Course, AEC Ability Enhancement Course; VAC Value Added Course: MD-Multidisciplinary Course 1 Credit (Theory)=15 Hours; 1 Credit (Practical)=30 Hours

Texts/References	- Gatrell, C. A. and Elliott, J. S. (2014): Geographies of Health: An Introduction. Wiley-Blackwell. 3rd Ed. - Meade, M. and EMNh, M. (2010): Medical Geography. Guilford Press. 3rd Ed. - EMNh, M., Root, D. E., Carrel, M. (2017): Health and medical geography. Guilford Press. 4th Ed. - Soja, E. W. (2010): Seeking Spatial Justice. University of Minnesota Press. Minneapolis. - Bonita, R., Beaglehole, R., Kjellström, T. & World Health Organization. (2006): Basic epidemiology. World Health Organization. 2nd Ed. - Grekousis, G. (2020): Spatial Analysis Methods and Practice: Describe-Explore-Explain Through GIS. Cambridge University Press. New York. 1st Ed. - Darmofal, D. (2015): Spatial Analysis for the Social Sciences. Cambridge University Press. New York. 1st Ed. - Reddy, S. K. (2019): Make Health in India: Reaching a Billion Plus. Orient Black Swan. 1st Ed.
Learning Outcomes	- Students will be introduced to the historical development, and key contributors. - Students would be able to understand about spatial diffusion of diseases. - They will learn the knowledge of improved sanitation and hygiene on health - Students should understand the National health policy, Major health and disease.

Course	B.Sc./B.A.FourthYear:Semester-VII			
CourseTitle	GGRMJ71:Research Methodology			
Category of Course	Major			
Credits&Hour of Teaching		Theory	Practical	Cumulative
	Credits	4		
	Hour of Teaching (Total)	60		
CourseObjectives	- To introduce the concept and significance of research geography. - To explain the philosophy and geography relations. - To introduce students about research process and design and data collection - To acquaint the students with data analysis, report writing and structure.			
CourseContent	The course content is divided into 4 Units.			
Units	CourseContent			Hr. of Teaching
I	Research in Geography: Concept and significance of research in geography; Production of geographical knowledge; Philosophy and Geography: Ontology, Epistemology, Methodology and Methods; Positivist and Non-Positivist Research			15
II	Research Process and Design: Research Process; Research designs : quantitative (experimental and survey), qualitative (ethnography, grounded theory, case study, phenomenological), mixed method (sequential, concurrent, and transformative); Ethics in Research.			15
III	Data Collection Methods: Primary data collection: questionnaires, observation, key information interviews, in-depth interviews, focus group discussion; Sources of data; Sampling and sample size determination; Quality control checks;			15
IV	Data Analysis and Dissemination: Data analysis: Quantitative (variables, scales, univariate, bivariate, and multivariate analysis), Qualitative (thematic analysis); Writing reports and research papers: structure, referencing, plagiarism.			15

SEC-Skill Enhancement Course, AEC Ability Enhancement Course; VAC Value Added Course: MD-Multidisciplinary Course 1 Credit (Theory)=15 Hours; 1 Credit (Practical)=30 Hours

Texts/References	- Ahuja, R. (2001): Research Methods. Rawat Publications, Jaipur and New Delhi. - Denzin, N.K. and Lincoln, Y.S., (eds.) (2000): Handbook of Qualitative Research. Thousand Oaks CA. Sage Publications. - Flowerdew, R. and Martin, D. (eds.) (1997): Methods in Human Geography. A Guide for Students Doing a Research Project. Longman, Harlow. - Kitchin, R. and Tate, N., (2001): Conducting Research into Human Geography: Theory, Methodology and Practice. Prentice-Hall, London. - Limb, M. (2001): Qualitative Methodologies for Geographers: Issue and Debates. Edward Arnold, London. - Mishra, H.N. & Singh, V.P. (2002): Research Methodology in Geography, Rawat Publications, Jaipur. - Neuman, Lawrence (2020): Social Research Methods: Qualitative and Quantitative Approach, Pearson - Rogers, A. and Viles, H.A. (2003): The Student's Companion to Geography. Blackwell Publishers, Oxford. Indian reprint available. - Silverman, D. (1993): Interpreting Qualitative Data: Methods for Analysing Talk, Text and Interaction. Sage Publications, London. - Tandon, B.C. (1997): Research Methodology in Social Sciences, Chaitanya Pub., Allahaba.
Learning Outcomes	- They would be able to enhance the learning on geographical research and its perspectives. - Students will be made aware about themes, approaches and recent developments. - They will be acquainted the knowledge of types of research designs and ethics. - Student should understand the data collection, analysis and presentation.

Course	B.Sc./B.A.FourthYear:Semester-VII			
CourseTitle	GGRMJ72:Environmental Studies			
Category of Course	Major			
Credits&Hour of Teaching		Theory	Practical	Cumulative
	Credits	4		
	Hour of Teaching (Total)	60		
CourseObjectives	- To introduce students to types and components of environment. - To explain the distribution and classification of natural resources. - To introduce students about environment issues and monitoring. - To acquaint the students with environmental conflicts and geopolitics.			
CourseContent	The course content is divided into 4 Units.			
Units	CourseContent			Hr.of Teaching
I	Environmental concepts: Meaning, types and components of the environment; Approaches to Environmental Studies; Ecosystem: components and classification; Geo-biochemical cycles: carbon, nitrogen and phosphorus.			15
II	Natural Resources and Biodiversity: Distribution and classification of Natural Resources: Land, Water, Forest and Energy; Biodiversity: Extinction and conservation; Biomes of the world.			15
III	Environmental problems and management: Environmental Issues in Tropical, Temperate and Polar Ecosystems; Pollution: air, water, soil and solid waste; Hazards: earthquake and flood; Environmental monitoring; Environmental education, Environmental policies and legislations.			15
IV	Trends in Environmental studies: Environmental conflicts; Environment and geopolitics; transnational movement; Environmental justice; Environmental geoinformatics; Environmental surveillance; Environmental mapping.			15

SEC-Skill Enhancement Course, AEC Ability Enhancement Course; VAC Value Added Course: MD-Multidisciplinary Course 1 Credit (Theory)=15 Hours; 1 Credit (Practical)=30 Hours

Texts/References	- Anjuneyulu, Y.(2004): Introduction to Environmental Science. B.S. Publications, Hyderabad. - Blaikie, P., Cannon, T. and Davis, I. (eds.) (2004): At Risk: Natural Hazards, Peoples Vulnerability and Disasters. Routledge, London. - Huggett, R.J. (1998): Fundamentals of Biogeography. Routledge, London. - Kayastha, S.L. and Kumra V.K. (1986): Environmental Studies. Tara Book Agency, Varanasi. - Kumaraswamy, K., Alagappa Moses., A and Vasanthy, M. (2004) Environmental Studies, Bharathidasan University, Tiruchirappalli. - Nag, P., Kumra, V.K. and Singh, J. (1990): Geography and Environmental Issues at Local, Regional and National Levels. (in 3 vols.), Concept Publishing Company, New Delhi. - Odum, E.P. (1975): Ecology. Rowman and Littlefield, Lanham. USA. - Rajagopalan, R. (2005): Environmental Studies. Oxford University Press, New Delhi. - Saxena, H.M., 2012: Environmental Studies, Rawat Publications, Jaipur. - UNEP (2007): The fourth Global Environment Outlook – environment for development (GEO-4). United Nations Environment Program.
Learning Outcomes	- Students would be able to learn types, components of environment and approaches. - Students will be made aware about distribution and biodiversity challenges. - They will get the knowledge of environmental issues, monitoring and policies. - Student should understand the environment geopolitics, justice and mapping.

Course	B.Sc./B.A.FourthYear:Semester-VII			
Course Title	GGRMJ73:Geographical Field Training and Report Writing			
Category of Course	Major			
Credits & Hour of Teaching		Theory	Practical	Cumulative
	Credits		2	
	Hour of Teaching (Total)		60	
Course Objectives	- To introduce students, the meaning and objectives of geographical field training. - To explain field as a laboratory of geography. - To understand various contents of field training. - To train students on field based report writing.			
Course Content	The course content is divided into 2 Units.			
Units	Course Content			Hr. of Teaching
I	Principles of Field Training: Meaning and objectives of field training; Field as a laboratory of geography; Bases of selection of field, Contents of field training: physical, social, economic and cultural; Field training techniques; Field training-based report writing			30
II	Field Visit: Students will visit any one of the following geographical regions of physical and human interest for field training: Himalayan Region, Vindhyan Plateau, Thar Desert and Aravali Hills, Coastal region, East and North-eastern India.			30

SEC-Skill Enhancement Course, AEC Ability Enhancement Course; VAC Value Added Course: MD-Multidisciplinary Course 1 Credit (Theory)=15 Hours; 1 Credit (Practical)=30 Hours

Texts/References	- Archer, J.E. and Dalton, T.H. (1968): Field Work in Geography. William Clowes and Sons Ltd. London and BeM Jles. - Bolton, T. and Newbury, P.A. (1968): Geography through Fieldwork. Blandford Press, London. - Jones, P.A. (1968): Field Work in Geography. Longmans, Green and Company Ltd., London and Harlow. - Lousenbury, J.F. and Aldrich, F.T. (1986): Introduction to Geographic Field Methods and Techniques. Charles E. Merrill Publishing Company, Columbus. - Pugh, J.C. (1975): Surveying for Field Scientists. Methuen and Company Ltd. London. - Parsons, Tony and Knight, Peter G. (2005): How to do your Dissertation in Geography and Related Disciplines. Routledge, London. 2nd Ed. - Kitchen, Rob and Tate, Nicholas J. (2009): Conducting Research into Human Geography: Theory, Methodology & Practice. Prentice Hall-Pearson, Harlow U.K. 2nd Ed. - Hay, Iain (ed.) (2005): Qualitative Research Methods in Human Geography. Oxford University Press, Melbourne. 2nd Ed. - Hay, Iain (ed.) (2004): Communicating in Geography and the Environmental Sciences. Oxford University Press, Melbourne. 2nd Ed. - Stoddard, Robert H. (1982): Field Techniques and Research Methods in Geography. Kendall/Hunt Pub. Dubuque IO.
Learning Outcomes	- Enhance students' understanding of the importance of fieldwork in geographical studies. - Increase awareness of key aspects and phenomena observed in the field. - Familiarize students with essential techniques used in fieldwork. - Enable students to effectively write reports based on field training experiences.

Course	B.Sc./B.A.FourthYear:Semester-VII			
CourseTitle	GGRMJ74A:Advanced Geographical Thought			
Category of Course	Major			
Credits&Hour of Teaching		Theory	Practical	Cumulative
	Credits	4		
	Hour of Teaching (Total)	60		
CourseObjectives	- To introduce students to the recent nature of geographical knowledge. - To explain new environmentalism and major approaches in geography. - To understand various paradigms and contributions of geographers. - To train students on contributions of Indian geographers and scholars.			
CourseContent	The course content is divided into 4 Units.			
Units	CourseContent			Hr. of Teaching
I	Basic Concepts: Nature of geographical knowledge; Man-environment interaction: new environmentalism; Concepts: space, place, environment, time And Spatial organization Culture and; cultural and scape.			15
II	Modern Approaches: Quantitative geography; Humanistic geography: phenomenology and existentialism; Contributions of Carl Sauer and Yi-Fu Tuan.			15
III	Contemporary Trends: Qualitative paradigm; Post-modernism and Post-structuralism; Social theory and De/reconstruction Welfare/Marxist; /Radical geography; Contributions of David Harvey; Feminist geography.			15
IV	Trends of Indian Geography: Colonial and Postcolonial geography; Contributions of Indian geographers; Impact of thoughts of Gandhi, Nehru and Ambedkar on Indian geography; Indian geography: issues and challenges.			15

SEC-Skill Enhancement Course, AEC Ability Enhancement Course; VAC Value Added Course: MD-Multidisciplinary Course 1 Credit (Theory)=15 Hours; 1 Credit (Practical)=30 Hours

Texts/References	- Daniels, P., Bradshaw, M., Shaw, D. and Sidaway, J. (2000): An Introduction to Human Geography. Issues for the 21st Century. Prentice Hall, London. - Dear, M. J. and Flusty, S. (2002): The Spaces of Postmodernity: Readings in Human Geography. Blackwell Publishers, Oxford. - Dikshit, R. D. (2004): Geographical Thought. A Critical History of Ideas. Prentice-Hall India, New Delhi. - Harvey, M. E. and Holly, P. B. (2002): Themes in Geographic Thought. Rawat Publications, New Delhi. - Hubbard, P., Kitchin, R., Bartley, B. and Fuller, D. (2002): Thinking Geographically: Space, Theory and Contemporary Human Geography. Continuum, London. - Johnston, R. J. and Sidaway, J. D. (2004): Geography and Geographers. 6th edition, Edward Arnold, London. - Kapur, A., (ed.) (2001): Indian Geography-Voice of Concern. Concept Publishing Company, New Delhi. - Martin, G., (2005): All Possible Worlds. A History of Geographical Ideas. 4th edition, Oxford University Press, New York. - Mathews, J. A., and Herbert, D. T. (eds.) (2004): Unifying Geography. Common Heritage, Shared Future. Routledge, London. - Soja, E. (1989): Post-modern Geographies. Verso Press, London. Reprinted 1997: Rawat Publications, Jaipur and New Delhi - Taylor, G. (ed.) (1953): Geography in the Twentieth Century. Mathuen and Company Ltd. And Company, London. - Tuan, Yi-Fu (1977): Space and Place. The Perspective of Experience. Edward Arnold, London.
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Learning Outcomes	- Enhance students' understanding of the evolving nature of geographical knowledge. - Familiarize students with various modern approaches in geography. - Acquaint students with contemporary trends and significant contributions in the field. - Enable students to understand trends in Indian geography
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Course	B.Sc./B.A.Fourth Year:Semester-VII			
Course Title	GGRMJ74B:Advances in Population Studies			
Category of Course	Major			
Credits&Hour of Teaching		Theory	Practical	Cumulative
	Credits	4		
	Hour of Teaching (Total)	60		
Course Objectives	- To introduce students about the meaning, scope and approaches to Population Studies. - To explain the factors, determinants and patterns & theories of population. - To understand composition, structure and factors of change in population. - To train students on population policies and contemporary issues.			
Course Content	The course content is divided into 4 Units.			
Units	Course Content			Hr. of Teaching
I	Basics of Population Studies: Meaning, Scope and approaches to Population studies; Sources of Population data with special reference to India; Demographic Dividend; Recent trends in population studies.			15
II	Population Distribution, Density and Growth: Factors, Determinants and Patterns; Theories of Population Growth: Malthusian theory and Demographic Transition Theory; Concept of Under, Over and Optimum Population; Population Resource Region.			15
III	Population Composition and Migration: Age-Sex, Literacy, Rural-Urban, OMJupation structure; Fertility; Mortality; Migration: Concept, Types, Determinants and Theories;			15
IV	Population policies and issues: Population Policies in Developed and Developing countries including India; Contemporary Issues in Population Geography-Ageing Population, Sex Ratio, Diseases, Epidemics and Population Problems; Population and Environment Interface; Population and Development: Human Development Index; New Research areas in Population Geography;			15

SEC-Skill Enhancement Course, AEC Ability Enhancement Course; VAC Value Added Course: MD-Multidisciplinary Course 1 Credit (Theory)=15 Hours; 1 Credit (Practical)=30 Hours

Texts/References	- Chandna, R.C. (2010) Population Geography, Kalyani Publisher, New Delhi - Hassan, M.I. (2020) Population Geography: A Systematic Exploration, Routledge India Pvt. Ltd, New Delhi - Tiwari, R.C. (2007) Geography of India, Prayag Pustak Bhawan, Allahabad - Barrett H.R., 1995: Population Geography, Oliver and Boyd. - Bhende A. and Kanitkar T., 2000: Principles of Population Studies, Himalaya Publishing House. - Chandna R.C. and Sidhu M.S., 1980: An Introduction to Population Geography, Kalyani Publishers - Jones, H.R., 2000: Population Geography, 3rd ed. Paul Chapman, London. - Lutz W., Warren C.S. and Scherbov S., 2004: The End of the World Population Growth in the 21st Century, Earthscan - Newbold K.B., 2009: Population Geography: Tools and Issues, Rowman and Littlefield Publishers - Pacione M., 1986: Population Geography: Progress and Prospect, Taylor and Francis. - Wilson M.G.A., 1968: Population Geography, Nelson.
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Learning Outcomes	- Students will enhance the learning on the basics of Population Geography. - They will make students aware about various factors affecting population growth. - They will be acquainted the knowledge of composition and structure of population. - Student should understand population issues and policies.
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Course	B.Sc./B.A.Fourth Year:Semester-VII			
Course Title	GGRMJ74C:Urban Geography			
Category of Course	Major			
Credits & Hour of Teaching		Theory	Practical	Cumulative
	Credits	4		
	Hour of Teaching (Total)	60		
Course Objectives	- To introduce students the nature, scope and recent trends in urban geography. - To explain the origin and evolution of urban settlements and their characteristics. - To understand the city size and urban structure models. - To train students on urban growth and contemporary issues in planning.			
Course Content	The course content is divided into 4 Units.			
Units	Course Content			Hr. of Teaching
I	Bases: Nature, scope and recent trends in urban geography; Urban population: characteristics, processes and trends; Methodology in urban studies; Origin and Evolution of Urban settlements: bazaar-based and colonial based; distribution of urban centres.			15
II	Characteristics: Characteristics of cities in different historical periods: pre-industrial, industrial and post-industrial; Functional classification of urban centres urban transportation; National Urban Information System.			15
III	Spatiality and Models: Size and spacing of cities: rank-size rule; law of the primate city; Urban hierarchies; Central place theory (Christaller and Loch); Urban land use and functional morphology: functional areas and peri-urban areas; Theories of urban structure: Burgess, Hoyt, Harris and Ullman, Mann, and White.			15
IV	Issues and Planning: Urban sprawl; urban problems: environmental, poverty, inequality, slums, transportation, housing, segregation and crime; Planned cities: Chandigarh and Jaipur; Concept of smart city in India; National Urban Policy; Master Plan: A case study of Varanasi.			15

SEC-Skill Enhancement Course, AEC Ability Enhancement Course; VAC Value Added Course: MD-Multidisciplinary Course 1 Credit (Theory)=15 Hours; 1 Credit (Practical)=30 Hours

Texts/References	- Bridge, B. and Watson, S. (Eds.) (2000): A Companion to the City. Blackwell, Oxford. - Carter, H. (1995): The Study of Urban Geography. 4th ed. Reprinted in 2000 by Rawat Publications, Jaipur and New Delhi. - Dutt, A. Allen, K. Nobel, G., Venugopal G. and Subbiah S. (eds.) (2003): Challenges to Asian Urbanisation in the 21st Century. Kluwer Academic Publishers, Dordrecht and London. - Hall, T. (2001): Urban Geography. 2nd edition. Routledge London. - Mayer, H. and Cohn, C. F. (1959): Readings in Urban Geography, University of Chicago Press, Chicago - Pacione, M. (2005): Urban Geography: A Global Perspective. Routledge, London and New York. - Ramachandran, R., (1991) Urbanization and Urban Systems in India. Oxford University Press, Delhi. - Singh, R. L. (1955): Banaras. A Study in Urban Geography. Nand Kishore and Brothers Banaras - Singh, H. H. (1972): Kanpur: A Study in Urban Geography, Indrasini Publications, Varanasi
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	10. Stanley,B.,Jack, W.andDonald,Z.(eds.)(2003):CitiesoftheWorld. RowmanandLittlefield, NewYorkandOxford. 11. Tripathi, V.K.,Jha,D.K.andHarshwardhanR.(eds.)(2018):AdvancesinUrbanStudiesinIndia.R.K.Books,NewDelhi.
Learning Outcomes	- Enhancingstudent'slearningonbasesandgrowthofUrbanGeography. - Students will be made aware about various factors affecting urban population. - They will be familiarized the knowledge of composition, urban structure and functions. - Studentshouldunderstandcontemporaryurbanpopulationissues, policiesandplanning.

Course	B.Sc./B.A.FourthYear:Semester-VII			
CourseTitle	GGRMJ74D:Regional Planning			
Category of Course	Major			
Credits&Hourof Teaching		Theory	Practical	Cumulative
	Credits	4		
	Hour of Teaching (Total)	60		
CourseObjectives	- Tointroducestudents theconcept,nature,andscopeofregionalplanning. - Todescribetheories,approachesandgeographicalissuesofregionalplanning. - Tounderstandtheevolutionandgrowthofregionalplanningin India - Totrainstudentsoncontemporaryregionaldevelopmentissuesandplanning.			
CourseContent	Thecoursecontentisdividedinto4Units.			
Units	CourseContent			Hr.of Teaching
I	ConceptualBackground: Concept,Natureandscopeofregionalplanning; Theories and approachesof regional planning, Concept of Planning region,PlanningregionsofIndia;Decentralizedplanning			15
II	Regional Development: Regional imbalances and in equalities inIndia;Regional planning and national development; Economic development and regional development; Role of agriculture, industry and infrastructure in regional development.			15
III	Regional Planning: Evolution of Regional Planning in India; Regional planning and National development; Planning for regional development,Integratedregionalplanning,Multi-level planning, Surveyforplanning:diagnostic,techno-Economic and regional			15
IV	PlanningandPolicyinIndia: RegionaldevelopmentstrategyinIndia;ChangingparadigmofplanninginIndia.Nationalandstatelevelplanning,Locallevelplanning;NationalCapitalRegion;PlanningforEasternUttarPradesh.NitiAayog.			15

SEC-SkillEnhancementCourse, AECAbilityEnhancementCourse; VACValueAdded Course: MD-MultidisciplinaryCourse1Credit(Theory)=15Hours;1Credit(Practical)=30Hours

Texts/References	- Chandna,R.C.(2000):RegionalPlanning:AComprehensiveText.KalyaniPublishers,NewDelhi. - Chaudhuri,J.R.(2001):AnIntroductiontoDevelopmentandRegionalPlanning withspecialreferencetoIndia.Orient Longman,Hyderabad. - Friedmann,J.andAlonso,W.(ed.)(1973):RegionalDevelopmentandPlanning.TheMITPress,Mass. - Kuklinski,A.R.(1972):GrowthPolesandGrowthCentresinRegionalPlanning.MoutonandCo.,Paris. - Misra,R.P.(ed.)(1992):RegionalPlanning:Concepts,Techniques,Policiesand CaseStudies.2nd Edition.Concept PublishingCompany,NewDelhi. - Pathak,C.R.(2003):SpatialStructureandProcessesofDevelopmentinIndia.RegionalScienceAssociation,Kolkata.
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	7. Sanyal, B.M. (2001): Decentralized Planning: Themes and Issues. Concept Publishing Company, New Delhi. 8. Sharma, P. V., Rao, V. L., and Pathak, C. R. (eds.) (2000): Sustainable Regional Development (with special reference to Andhra Pradesh). Regional Science Association, Kolkata and School of Economics, Andhra University, Vishakhapatnam. 9. Sundaram, K. V. (1997): Decentralized Multilevel Planning: Principles and Practice (Asian and African Experiences). Concept Publishing Company, New Delhi. 10. Sundaram, K. V. (2004): The Trodden Path: Essay on Regional and Micro Level Planning. Anaanya Publications, New Delhi.
Learning Outcomes	• Deepen students' understanding of the fundamental aspects of regional planning. • Increase awareness of the factors and issues related to regional imbalances. • Familiarize students with the evolution of regional planning in India. • Students should understand contemporary regional development strategy.

Course	B.Sc./B.A. Fourth Year: Semester-VII			
Course Title	GGRMJ74E: Advanced Geographical Information System			
Category of Course	Major			
Credits & Hour of Teaching		Theory	Practical	Cumulative
	Credits	4		
	Hour of Teaching (Total)	60		
Course Objectives	• To introduce students, the basics of GIS and its developments. • To describe and explain data types and models, structures and data quality. • To develop students' understanding of spatial data analysis. • To train students on recent advances and its applications.			
Course Content	The course content is divided into 4 Units.			
Units	Course Content			Hr. of Teaching
I	Concepts of Geographical Information System: Development of Geographical Information System, Terminologies; Data types, characteristics and sources; Data Models-Conceptual, Logical, and Physical; Data Structures-Hierarchical, Relational and Object oriented			15
II	Data Base and Standards: Data Base Management System (DBMS); Data Quality: meaning, measures, components and management; Geographical Information System Data Standards: meaning, classification, and components.			15
III	Spatial Data Analysis: Raster Data Analysis: Local; neighborhood and regional operations-Overlay Analysis, Map Algebra; Vector Data Analysis: Concept of Topology and Topological Analysis; Proximity Analysis: Buffering, Thiessen Polygon; Network Analysis; Multi-Criteria Analysis; Data Interpolation: IDW and Kriging.			15
IV	Recent Advances and Applications: Web GIS, Global Positioning System (GPS) and Mobile mapping; Machine learning; Artificial Intelligence (AI); Geographical Information System Software and programming.; Geographical Information System Applications: Land Use Planning, Site Suitability Analysis, Water Resource Management, Disaster Management and Mitigation.			15

SEC-Skill Enhancement Course, AEC Ability Enhancement Course; VAC Value Added Course; MD-Multidisciplinary Course 1 Credit (Theory)=15 Hours; 1 Credit (Practical)=30 Hours

Texts/References	1. Bolstad, P. (2005). GIS Fundamentals: A First Text On Geographic Information Systems. Elsevier Press. 2. Burrough, P. A. and M. N. Donnell, R. (1998): Principles of Geographic Information Systems. Oxford University Press, Oxford. 3. Demers, M. N. (2000): Fundamentals of Geographic Information Systems. John Wiley and Sons, Singapore. 4. Demers, M. N. (2008). Fundamentals of Geographic Information Systems. John Wiley & Sons. 5. Fu, P., & Sun, J. (2010). Web GIS: Principles and Applications. Esri Press. 6. Hofmann-Wellenhof, B., Lichtenegger, H., & Wasle, E. (2007). GNSS – Global Navigation Satellite 7. Korte, G. M. (2002): The GIS Book. On Word Press: Thomson Learning, New York and Singapore. 8. Maguire, D. J., Michael F. G. and David W. R. (1999): Geographical Information Systems: Principles and Application. Geo Information International, Vol. 2, Longman Publication., New York. 9. Martin, D. (1996): Geographic Information Systems: Socioeconomic Implications. Routledge, London. 10. Michael F. G. and Karan K. K. (ed.) (1990): Introduction to GIS. NCGIA, Santa Barbara, California. 11. Ripple, W. J. (ed.) (1989): Fundamentals of Geographic Information Systems: A Compendium. ASPRS/ACSM, Falls Church. 12. Star, J. and Estes, J. (1990): Geographic Information Systems – An Introduction. Prentice-Hall, Englewood Cliffs, New Jersey. 13. Worboys, M. F. (1995): GIS: A Computing Perspective. Taylor and Francis, London.
Learning Outcomes	• Enhance students' understanding of the fundamental aspects of GIS. • Familiarize students with key terminologies, data types, sources, and data models in GIS • Students should be able to understand spatial data analysis and GIS Data Standards. • Students should understand nature of spatial data, software and applications

Course	B.Sc./B.A. Fourth Year: Semester-VII			
Course Title	GGRMJ74F: Rural and Urban Planning			
Category of Course	Major			
Credits & Hour of Teaching		Theory	Practical	Cumulative
	Credits	4		
	Hour of Teaching (Total)	60		
Course Objectives	• To introduce students about the basics and concepts of rural and urban planning. • To describe and explain approaches of rural planning and infrastructure. • To understand and comprehend techniques of urban planning and land use models. • To train students on urban planning approaches, systems and case studies.			
Course Content	The course content is divided into 4 Units.			
Units	Course Content			Hr. of Teaching
I	Concept of Rural Planning: Concept and approaches of rural planning; Policies and programmes of rural development; Basic infrastructures for rural development; Decentralized planning; Social impact assessment analysis for rural planning; Urban-rural linkage approach to planning.			15
II	Approaches to Rural Planning: Land use planning: concepts, principles and classification; Carrying capacity of land; Agricultural efficiency; Rural land patterns and issues of land acquisition in India; Rural infrastructure; Microfinance and Micro Enterprises; Local level planning; PURA; Village Information System			15

III	Concept of Urban Planning: Concept and approaches of urban planning; Techniques and methods of urban planning; Urban land use: patterns and models; Urban renewal and redevelopment; Urban governance: Urban infrastructure; Master plan: A case study of Varanasi.	15
IV	Approaches to Urban Planning: Garden cities, City as an organism, New and satellite town, global city, sustainable smart cities; Transportation planning: Mass transit system; Heritage conservation; Urban planning: A case study of Chandigarh; National Urban Information system.	15

SEC-Skill Enhancement Course, AEC Ability Enhancement Course; VAC Value Added Course: MD-Multidisciplinary Course 1 Credit (Theory)=15 Hours; 1 Credit (Practical)=30 Hours

Texts/References	1. Bhat, L.S. (1976): Micro Level Planning in India. K.B. Pub. New Delhi. 2. Desai, A.R. (1990): Rural Development. Popular Prakashan, Bombay. 3. Desai, V. and Potter, R. (eds.) (2002): The Arnold Companion of Development Studies. Arnold, London. Found, W.C. (1982): Theoretical Approach to Rural Land Use Pattern. Methuen and Company Ltd., London. 4. Hall, P. (1992): Urban and Regional Planning. Routledge, London. 5. Jr. Chapin S. F., Kaiser, E. J. Godschalk, D. R. (1995): Rural and Urban Land-use Planning. University of Illinois Press Urban and Chicago 6. Krishnamurthy, J. (2000): Rural Development. Problems and Prospects. Rawat Publications, Jaipur. 7. Lejonhud, K. (2003): Indian Villages in Transformation: A Longitudinal Study of Three Villages in Uttar Pradesh. Karlstad University Studies, Karlstad. 8. Misra, R. P. and Achyutha, R. N. (1998): Micro-Level Rural Planning: Principles, Methods and Case Studies. Concept Publishing Company, New Delhi. 9. Misra, R. P. and Misra, K. (eds.) (1998): Million Cities of India: Sustainable Development Foundation, New Delhi. Banaras Hindu University Institute of Science, Department of Geography M.Sc./M.A. Two-Year Semester Course Outline, 2019-2028 10. Sundaram, K.V. (1977): Urban and Regional Planning in India. Vikas, New Delhi. Varanasi, Pub
Learning Outcomes	• Enhance students' understanding of fundamental concepts in rural planning. • Increase awareness of various approaches to rural planning. • Familiarize students with urban planning practices and urban land use models. • Enable students to understand planning practices in India as well as in other contexts.

Course	B.Sc./B.A. Fourth Year: Semester-VII			
Course Title	GGRMJ74G: Geography of Migration and Diaspora			
Category of Course	Major			
Credits & Hour of Teaching		Theory	Practical	Cumulative
	Credits	4		
	Hour of Teaching (Total)	60		
Course Objectives	• To introduce students about meaning, definitions and types of Migration. • To describe and explain Spatial dimension of Internal Migration in India. • To understand and comprehend determinants of International Migration. • To train students on Indian diaspora: identity, ethnicity and citizenship.			
Course Content	The course content is divided into 4 Units.			
Units	Course Content			Hr. of Teaching
I	Basic Concepts: Meaning, Definitions and Types of Migration and Diaspora; Causes and Consequences of Migration; Laws and Theories			15

	of Migration; Sources and Data of Migration and Diaspora.	
II	Internal Migration: Spatial Dimension of Internal Migration in India: Inter-State and Intra-State; Trends and Patterns of Internal Migration; Reasons, Streams and Consequences of Internal Migration.	15
III	International Migration and Diaspora: Determinants of International Migration; Globalization, International Migration and Diaspora; Illegal Migration: Causes and Consequences; Role of Organizations: International Organization for Migration (IOM), United Nations High Commissioner for Refugees (UNHCR)	15
IV	Indian Diaspora: Indian Diaspora across Continents and Countries; Indian Diaspora: Identity, Ethnicity, Citizenship, People of Indian Origin, Remittances; Challenges faced by Indian Diaspora.	15

SEC-Skill Enhancement Course, AEC Ability Enhancement Course; VAC Value Added

Course: MD-Multidisciplinary Course 1 Credit (Theory)=15 Hours; 1 Credit (Practical)=30 Hours

Texts/References	1. Cohen, Robin 1997. Global Diaspora: An Introduction. London: UCL Press. 2. Jana Evans Braziel and Anita Mannur. 2003. "Nation, Migration, Globalization: Points of Contention in Diaspora Studies." In: Theorizing Diaspora. Ed. Braziel and Mannur. Oxford: Blackwell, pp. 1-10. 3. Kim D. Butler. 2002. "Defining Diaspora, Refining a Discourse," Diaspora 10, 189-219. 4. Gilroy, Paul. 2003. "The Black Atlantic as a Counter Culture of Modernity." In: Theorizing Diaspora. Ed. Braziel and Mannur. Oxford: Blackwell, pp. 49-80. 5. Stephen Castles and Mark J. Miller. 1998. The Age of Migration: International Populations Movements in the Modern World. London: Macmillan. 6. Hall, Stuart. 2003. "Cultural Identity and Diaspora." In: Theorizing Diaspora. Ed. Braziel and Mannur. Oxford: Blackwell, pp. 233-246. 7. Thapan, Meenakshi (ed.). 2005. Transnational Migration and the Politics of Identity. New Delhi: Sage. Textbooks 8. Vertovec, Steven and Robin Cohen (eds.). 1999. Migration, Diaspora and Transnationalism. London: Edward Elgar. 9. Hegde, R. & Sahoo, A.K. (eds.). 2018. Routledge Handbook of the Indian Diaspora. New York: Routledge. 10. Cohen, Robin. (eds.). 2018. Routledge Handbook of Diaspora Studies. London: Routledge Publications.
Learning Outcomes	• Enhance students' understanding of the causes and consequences of migration. • Increase awareness of trends and patterns in internal migration. • Familiarize students with international migration and diaspora. • Enable students to understand the characteristics of the Indian diaspora.

Course	B.Sc./B.A. Fourth Year: Semester-VII			
Course Title	GGRMN71: Research Methodology			
Category of Course	Minor			
Credits & Hour of Teaching		Theory	Practical	Cumulative
	Credits	4		
	Hour of Teaching (Total)	60		
Course Objectives	• To introduce the concept and significance of research geography. • To explain the philosophy and geography relations. • To introduce students about research process and design and data collection.			

	To acquaint the students with data analysis, report writing and structure.	
Course Content	The course content is divided into 4 Units.	
Units	Course Content	Hr. of Teaching
I	Research in Geography: Concept and significance of research in geography; Production of geographical knowledge; Philosophy and Geography: Ontology, Epistemology, Methodology and Methods; Positivist and Non-Positivist Research	15
II	Research Process and Design: Research Process; Research designs: quantitative (experimental and survey), qualitative (ethnography, grounded theory, case study, phenomenological), mixed method (sequential, concurrent, and transformative); Ethics in Research	15
III	Data Collection Methods: Primary data collection: questionnaires, observation, key information interviews, in-depth interviews, focus group discussion; Sources of secondary data; Sampling and sample size determination; Quality control checks; Data cleaning.	15
IV	Data Analysis and Dissemination: Data analysis: Quantitative (variables, scales, univariate, bivariate, and multivariate analysis), Qualitative (thematic analysis); Writing reports and research papers: structure, referencing, plagiarism.	15

SEC-Skill Enhancement Course, AEC Ability Enhancement Course; VAC Value Added Course; MD-Multidisciplinary Course 1 Credit (Theory)=15 Hours; 1 Credit (Practical)=30 Hours

Texts/References	- Ahuja, R. (2001): Research Methods. Rawat Publications, Jaipur and New Delhi. - Denzin, N. K. and Lincoln, Y. S., (eds.) (2000): Handbook of Qualitative Research. Thousand Oaks CA. Sage Publications. - Flowerdew, R. and Martin, D. (eds.) (1997): Methods in Human Geography. A Guide for Students Doing a Research Project. Longman, Harlow. - Kitchin, R. and Tate, N., (2001): Conducting Research into Human Geography: Theory. Methodology and Practice. Prentice-Hall, London. - Limb, M. (2001): Qualitative Methodologies for Geographers: Issue and Debates. Edward Arnold, London. - Mishra, H. N. & Singh, V. P. (2002): Research Methodology in Geography, Rawat Publications, Jaipur. - Neuman, Lawrence (2020): Social Research Methods: Qualitative and Quantitative Approach, Pearson - Rogers, A. and Viles, H. A. (2003): The Student's Companion to Geography. Blackwell Publishers, Oxford. Indian reprint available. - Silverman, D. (1993): Interpreting Qualitative Data: Methods for Analysing Talk, Text and Interaction. Sage Publications, London. - Tandon, B. C. (1997): Research Methodology in Social Sciences, Chaitanya Pub., Allahabad.
Learning Outcomes	- They would be able to enhance the learning on geographical research and its perspectives. - Students will be made aware about themes, approaches and recent developments. - They will be acquainted the knowledge of types of research designs and ethics. - Student should understand the data collection, analysis and presentation..

Course	B.Sc./B.A.FourthYear:Semester-VII			
Course Title	GGRIN71: Internship			
Category of Course	Internship-2			
Credits & Hour of Teaching		Theory	Practical	Cumulative
	Credits	2		
	Hour of Teaching (Total)	30		

Course	B.Sc./B.A.FourthYear:Semester-VIII			
Course Title	GGRMJ83C: Advances in Urban Planning			
Category of Course	Major			
Credits & Hour of Teaching		Theory	Practical	Cumulative
	Credits	4		
	Hour of Teaching (Total)	60		
Course Objectives	• To introduce students about forms of urbanization and recent urban growth. • To describe and explain evolution of urban planning, paradigms and real estate factors. • To understand and comprehend urban governance and management practices. • To train students on case studies of urban planning and regeneration and new projects.			
Course Content	The course content is divided into 4 Units.			
Units	Course Content			Hr. of Teaching
I	Conceptual and Theoretical Background: Changing forms of urbanisation; Growth of cities; Stages of urban development; Evolution of urban planning; Objects of urban planning, Economic justification for urban planning; Principles of urban planning			15
II	Urban Transitions: Urban revolution; Fordist and Post-Fordist city; Land economics and urban restructuring; Paradigms in urban planning; Post modern urbanism; New urbanism; Cities and globalization; Impact of large urban projects; Real estate and cities; Right to the city;			15
III	Governance and Management: Inclusive city challenge; Local urban governance; Climate change and urban resilience; Urban ecology and sustainable urban development; Measuring sustainability; Smart urban growth; Urban ecology in sustainable urban development; Problems and remedies of urbanization; Local Agenda 21 and urban management;			15
IV	Cases of Urban Practices in India: Nature of urban regeneration and renewal; Spatiality of the slum (re)development; Transport :transit -oriented development; and Peri-urban growth; Urban participatory process under the CAA; Urban planning reforms			15

SEC-Skill Enhancement Course, AEC Ability Enhancement Course; VAC Value Added Course: MD-Multidisciplinary Course 1 Credit (Theory)=15 Hours; 1 Credit (Practical)=30 Hours

Texts/References	1. Bansal, S.C. (2010): Urban Geography, Meenakshi Prakashan, Meerut 2. Baud, I.S.A. and Wit, J. de (2008): New Forms of Urban Governance in India: Shifts, Models, Networks and Contestations, Sage, New Delhi 3. Cadwallader, Martin (1986): Urban Geography, Prentice Hall, New Jersey 4. Dickinson, R.E. (1964): City and Region, Routledge, London 5. Das, Amiya Kumar (2007): Urban Planning in India, Rawat Publications, Jaipur 6. Kumar, Ashok, Sanjeev Vidyarthi and Poonam Prakash (2020): City Planning in India, Routledge, New Delhi
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	7. Gavsker, Kapil Kumar (2012): India's Urban Development-Revitalizing Planning in Neoliberal Times & Integrating Governance, Lambert Academic Publishing, Germany 8. Kundu, A (1992): Urban Development and Urban Research in India, Khanna Publication, New Delhi 9. Mayer H.M. and Kohn, C.F. (eds.) (1958): Readings in Urban Geography, University of Chicago Press, Chicago 10. Prasad, Sheela and Kapil Kumar Gavsker (2016): 'The New 'Love' Story of the Taj Mahal- Urban Planning in the Age of Heritage Tourism in Agra', Economic and Political Weekly, LI(5):40-48 11. Ramachandran, R. (1989): Urbanisation and Urban Systems in India, Oxford, New Delhi. 12. Singh, K. and Steinberg, F. (eds.) (1987), Urban India in Crisis, New Age International, New Delhi
Learning Outcomes	- Enhancing student's learning on new forms of urbanization and its patterns. - To make students aware about urban growth and urban planning. - Familiarize students of local urban governance and concept to inclusivity. - Student should understand urban regeneration, renewal and peri-urban growth management.

Course	B.Sc./B.A. Fourth Year: Semester-VIII			
Course Title	GGRMJ83D: Industrial Geography			
Category of Course	Major			
Credits & Hour of Teaching		Theory	Practical	Cumulative
	Credits	4		
	Hour of Teaching (Total)	60		
Course Objectives	- To introduce students about meaning and scope of industrial geography. - To describe and explain factors of industrial location; Linkage in industries. - To understand localization and distribution of iron & steel, cotton textile and sugar industries. - To train students on industrial development and policies..			
Course Content	The course content is divided into 4 Units.			
Units	Course Content			Hr. of Teaching
I	Basic Concepts: Meaning and scope of industrial geography; Recent trends in industrial geography; Classification of industries; Concept of industrialization, industrial complex and industrial estate; Role of industries in regional development.			15
II	Location Factors and Theories: Factors of industrial location; Linkage in industries; Theories of industrial location: Weber, Hoover, Lösch and Smith.			15
III	The spatial arrangement of Industries: Localization and distribution of iron & steel, cotton textile and sugar industries; Bases of identification of industrial regions; Industrial region of the U.S.A. and West Europe.			15
IV	Industrialization in India: Industrial development and policies; Industrial regions and complexes; Impact of globalization on industries; Problems of industrialization; Environmental impact of industrialization; Tourism as an industry; Tourism in Uttar Pradesh. Special economic zones.			15

SEC-Skill Enhancement Course, AEC Ability Enhancement Course; VAC Value Added Course: MD-Multidisciplinary Course 1 Credit (Theory)=15 Hours; 1 Credit (Practical)=30 Hours

Texts/References	1. Alexanderson, C. (1967): Geography of Manufacturing. Prentice-Hall of India, New Delhi 2. Behari, Bepin, 1976. Rural Industrialization in India. Vikas Publishing House Pvt. Ltd. New Delhi, 3. Clark, G.L., Gertler, M.S. and Feldman, M.P. (eds.) 2000. The Oxford Handbook of Economic Geography. Oxford University Press, USA. 4. Coe, N. 2007. Economic Geography: A Contemporary Introduction. Blackwell Publishers, Inc., Massachusetts. 5. Dicken, P. and Lloyd, P. 1990/1977. Location in Space. New York: Harper & Row 6. Hoover, E.M. (1948): Location and Space Economy. MNGraw-Hill, New York. 7. Hudson, R. 2005. Economic Geographies: Circuits, Flows and Spaces. Sage Publications, London. 8. Mackinnon, D. and Cumbers, A. 2007. Introduction to Economic Geography: Globalization, Uneven Development and Place. Prentice Hall, Inc., New York 9. Miller, E. (1962): Geography of Manufacturing. Prentice-Hall, Englewood Cliffs, New Jersey. 10. Riley R.C. 1973. Industrial Geography, Chatto and Windus Ltd. London. 11. Sharma, V.N. 2001. Industrial Development and Planning in India. Radha Publications, New Delhi. 12. Sheppard, E. and Barnes, T.J. (eds.) 2002. A Companion to Economic Geography. Blackwell Publishers, Inc., Massachusetts.
Learning Outcomes	• Enhancing student's learning on growth of industrial geography. • To make students aware factors and major theories of industrial location. • Familiarize students' about localization and distribution of major industries. • Student should understand nature of industrialization in India.

Course	B.Sc./B.A. Fourth Year: Semester-VIII			
Course Title	GGRMJ83E: Gender Geography			
Category of Course	Major			
Credits & Hour of Teaching		Theory	Practical	Cumulative
	Credits	4		
	Hour of Teaching (Total)	60		
Course Objectives	• To introduce students about meaning and scope of gender geography. • To describe and explain major approaches to women and development. • To understand socio-spatial construction of gender • To train students on mapping women's lives and violence against women.			
Course Content	The course content is divided into 4 Units.			
Units	Course Content			Hr. of Teaching
I	Conceptualizing Gender: Meaning and Scope of Gender Geography, Basic Concepts: sex, gender, sexuality, power, gender inequality; Feminism; Waves of feminism; Feminism in India; History of Indian Women's Movement.			15
II	Gender and Development: Approaches to women and development: WID, WAD & GAD; Gender empowerment measures (GEM); Sustainable development and gender, Global gender gap (GGP); Women and natural resource management (NRM), Gender mainstreaming; Gender budgeting.			15
III	Gender, Identity and Space: Socio-spatial construction of gender: class, caste, religion, sexuality and age; Public and private spaces: gender and work, gender and migration, sexual harassment: domestic and at work places; Legislative			15

	measurements: Domestic Violence act (2005), Protection of women against sexual harassment at workplace act(2013), Criminal law (Amendment)act-2013	
IV	Gendered Nature of House work; Girls Education; Women's Employment; Feminization of Poverty; Gender Violence; Women in Decision Making; HumanTrafficking.	15

SEC-SkillEnhancementCourse, AECAbilityEnhancementCourse; VACValueAdded

Course: MD-MultidisciplinaryCourse1Credit(Theory)=15Hours;1Credit(Practical)=30Hours

Texts/References	1. Bedi,Tarini(2006).FeministTheoryandtheRight-Wing: ShivSena Women Mobilize Mumbai.JournalofInternationalWomen'sStudies,7(4),51-68, available at :http://vc.bridgew.edu/jiws/vol7/iss4/4 2. Bhatt,ArchanaPathak(2008).TheSitaSyndrome:ExaminingtheCommunicative Aspects of DomesticViolence from a South Asian Perspective, Journal of International Women'sStudies,9(3),155-173. Available at: http://vc.bridge.w.edu/jiws/vol9/iss3/9 3. Hall, Jennifer (2002): The Next Generation: Can There Be a Feminist Geography without Gender? The Great Lakes Geographer,9 (1),19-27, available at: http://geography.uwo.ca/research/great_lakes_geographer/GLG volume9/Hall.pdf 4. Hanson,Susan(2005):Geographyandfeminism,WorldsinCollision,AnnalsoftheAssociationofAmericanGeographers,82 (4). 5. Harris,Ashleigh(2003):FromSuffragisttoApologist:TheLossofFeministPoliticsina PoliticallyCorrectPatriarchy.JournalofInternationalWomen's Studies,4(2),91-99, available at: http://vc.bridgew.edu/jiws/vol4/iss2/8 6. Haas, Hein de (2008): Migration and Development: A Theoretical Perspective, Working Paper, 09, International MigrationInstitute, JamesMartin 21st Century School. 7. MNDowell,Linda(1999):Gender,identityandplace:understandingfeministgeographies.PolityPress,Minnesota. 8. Mitra, Aditi; Bhatia, Manjeet; and Chatterjee, Sobha (2013): Perspectives on Women's Studies from India: Strengths, Struggle sand Implications for Programs in the U.S.,JournalofInternationalWomen'sStudies,14(3),194-209, available at :http://vc.bridgew.edu/jiws/vol14/iss3/14 9. Pamela,MossandAl-Hindi Karen Falconer An Introduction: Feminisms ,Geographies, Knowledges, 10. Sen, Samita (2000): Toward a Feminist Politics? The Indian Women's Movement in Historical Perspective, policy researchreport ongender and development, The World Bank, Working Paper Series No. 9, Seager, Joni (2009): The Penguin Atlas of women in theWorld,FourthEdition,Brighton:MyraidEditions,PenguinGroup
Learning Outcomes	• Enhancing student's learning on growth of gender geography. • Students will be made aware about the factors affecting women's lives. • They will be able to know about gender development approaches and programmes. • Student should understand gendered nature of house work and its implications.

Course	B.Sc./B.A.FourthYear:Semester-VIII			
Course Title	GGRMJ83F:Agriculture Geography			
Category of Course	Major			
Credits & Hour of Teaching		Theory	Practical	Cumulative
	Credits	4		
	Hour of Teaching (Total)	60		
Course Objectives	• To introduce students about meaning and scope of agriculture geography. • To describe and explain crop concentration and crop diversification. • To understand Concepts of agricultural land-use and carrying capacity and models. • To train students on Agro-climatic regions of India; Reforms in Indian agriculture.			
Course Content	The course content is divided into 4 Units.			
Units	Course Content			Hr. of Teaching
I	Basic Concepts: Meaning and scope of agricultural geography; Approaches to agricultural geography; Physical, cultural and institutional factors affecting agricultural distribution and productivity.			15
II	Agricultural Patterns: Crop concentration and crop diversification; Delineation of crop combination regions; Agricultural regions of the world; Agricultural Systems (subsistence, plantation and commercial farming).			15
III	Land use and Productivity: Concepts of agricultural land-use and carrying capacity; Model of Agricultural Land use (Von Thunen); Land use pattern and planning; Measures of agricultural efficiency and productivity.			15
IV	Indian Agriculture: Agro-climatic regions of India; Reforms in Indian agriculture with special reference to land and institutional reforms; Agricultural revolutions in India; Organic and contract farming; Agricultural planning and policies in India.			15

SEC-Skill Enhancement Course, AEC Ability Enhancement Course; VAC Value Added Course: MD-Multidisciplinary Course 1 Credit (Theory)=15 Hours; 1 Credit (Practical)=30 Hours

Texts/References	1. Dumont, R. (1970): Types of Rural Economy: Studies in World Agriculture, Douglas Manin, London and Methuen 2. Gregor, H.P. (1970): Geography of Agriculture. Prentice-Hall, New York. 3. Husain, M. (1996): Systematic Agricultural Geography, Rawat Publications, Jaipur. 4. Misra, R.P. (1967): Diffusion of Agricultural Innovations, University of Mysore, Mysore. 5. Mohammad, A. (1978): Studies in Agricultural Geography, Rajesh Publications, New Delhi 6. Morgan, W.B. and Norton, R.J.C. (1971): Agricultural Geography. Methuen, London. 7. Sauer, O.C. (1969): Agricultural Origins and Dispersals. MIT Press, Cambridge. 8. Shafi, M. (2006): Agricultural Geography, Pearson Education, New Delhi. 9. Sen, Sudhir (1975): Reaping the Green Revolution. Tata McGraw-Hill, New Delhi 10. Shafi, M. (2000): Agricultural Geography of South Asia, MN Millan, Delhi 11. Singh, B.B. (1979): <i>Krishi Bhugol</i>. Tara Publications, Varanasi. 12. Singh, J. and Dhillon, S.S. (2000): Agricultural Geography. Tata McGraw Hill, New Delhi. 13. Singh, S. (1994): Agricultural Development in India: A Regional Analysis, Kaushal Publications, Shillong. 14. Symons, L. (1967): Agricultural Geography. George Bell and Sons, London. 15. Tarrant J.R. (1974): Agricultural Geography. John Wiley and Sons, New York.
Learning Outcomes	• Enhancing student's learning on nature and approaches of agriculture geography • Students will be made aware to agricultural patterns and their measurements. • Students will be able to develop the knowledge of land use and productivity. • Student should understand agricultural regions and their characteristics..

Course	B.Sc./B.A.Fourth Year:Semester-VIII			
Course Title	GGRMJ83G:Environmental Hazards and Disaster Risk Reduction			
Category of Course	Major			
Credits&Hour of Teaching		Theory	Practical	Cumulative
	Credits	4		
	Hour of Teaching (Total)	60		
Course Objectives	• To introduce students about meaning and scope of environmental hazards • To describe and explain risk reduction strategies. • To understand Concepts of hazard identification and mapping. • To train students on international and national initiatives.			
Course Content	The course content is divided into 4 Units.			
Units	Course Content			Hr. of Teaching
I	Understanding Environmental Hazards: Concepts and Definition of Environmental Hazards, Disaster, Risk, and Resilience, Bases of Classification and Types of Hazards, Characteristics, spatio-temporal distribution, impact and risk reduction measures of Major geological hazards: Volcanic eruption, Earthquakes, Tsunami and Landslides; Major hydro-meteorological hazards: Floods, Drought, Cyclones, Cloud bursts.			15
II	Disaster Risk Reduction Process: Paradigm and approaches in disaster management, Concepts of Disaster Risk Reduction, Disaster Management Cycle, Pre-disaster stage-preparedness, prevention, mitigation; On disaster stage-response, Post-disaster stage-Relief, Recovery, Rehabilitation.			15
III	Techniques and Methods: Process of hazard identification and mapping, Indicators and factors of vulnerability, Risk assessment process			15
IV	International and National initiatives: UN and its Programmes: International Decade of Natural Disaster Reduction (IDNDR), International Strategy for Disaster Risk Reduction (ISDR), Hyogo Framework for Action (HFA) – 2005, Major hazards profile of India: Flood, drought, earthquake, tsunami, cyclones; Institutions: National Disaster Management Authority (NDMA), National Institute of Disaster Management (NIDM), Policy: Disaster management act 2005, Disaster management system in India.			15

SEC-Skill Enhancement Course, AEC Ability Enhancement Course; VAC Value Added

Course: MD-Multidisciplinary Course 1 Credit (Theory)=15 Hours; 1 Credit (Practical)=30 Hours

Texts/References	1. Alexander, D. (1993): Natural Disasters, Springer, Berlin 2. Bolt, B.A. et al. (ed.) (1950): Geological Hazards, Springer, Verlag, New York. 3. Burton, I. et al. (1978) Environmental Hazards, O.P.O., New York. 4. Goudie, A. (1990): Geomorphological Techniques, Unwin Hyman, London 5. Hart, M.G. (1986): Geomorphology: Pure and Applied, George Allen and Unwin, London 6. Kusky, T. (2012): Encyclopedia of the Hazardous Earth, Viva Books, New Delhi 7. Petak, W.J. and Atkinson, A.D. (1982): Natural Hazards Risk Assessment and Public Policy, Springer, Verlag, New York. 8. Roy, P.S. et al. (2000): Natural Disaster and their Mitigation, ITC Publication. 9. Valdiya, K.S. (1987): Environmental Geology, Tata McGraw Hill, New Delhi 10. White, G.F. (ed) (1974): Natural Hazards, Local, National and Global, Oxford University Press, London.
Learning Outcomes	• Enhancing student's learning on nature of hazards and their detection. • Students will be able to develop the knowledge of approaches in disaster management. • Students will be able to learn the knowledge of technique and methods of disaster management. • Students should understand major programmes, policies and case studies.

Course Title	GGRMJ83H: Ecotourism & Geoheritage			
Category of Course	Major			
Credits & Hour of Teaching		Theory	Practical	Cumulative
	Credits	4		
	Hour of Teaching (Total)	60		
Course Objectives	• To introduce students the nature and scope of Geography of eco-tourism. • To describe and explain tourism multiplier and forecasting methods. • To understand and comprehend cultural heritage-based tourism. • To train students on tourism and ecotourism in India & characteristics of tourism plan.			
Course Content	The course content is divided into 4 Units.			
Units	Course Content			Hr. of Teaching
I	Conceptual: Nature, and Scope of Geography of eco-tourism; Data sources; Typology of Tourism: developed and developing world; Tourism as an industry; Drivers of Tourism; Tourism and Regional Economy; Concept of heritage and its types.			15
II	System of Tourism and Impacts; Tourism Multiplier and Forecasting Methods; Trends and Pattern of tourism in the World; Models of Ecotourism; Impacts of Tourism-Socio-cultural problems and Environmental degradation.			15
III	Concept of Historic Urban Landscapes; Concept and Classification of heritage; Built heritage conservation; Cultural heritage-based tourism; Major National Parks, Beaches and Resorts of India; Built Heritage Conservation; Cultural Heritage Based Tourism; Legislation and Acts; Sustainable Tourism Development Planning			15
IV	Policy and Planning for Heritage; Tourism in India; Characteristics of Tourism Plan; Cases of Cultural Heritage: Conservation and Revitalization; Legislation and Acts; Sustainable Tourism Development Planning.			15

SEC-Skill Enhancement Course, AEC Ability Enhancement Course; VAC Value Added

Course: MD-Multidisciplinary Course 1 Credit (Theory)=15 Hours; 1 Credit (Practical)=30 Hours

Texts/References	1. Bhatia A.K. (1996): Tourism Development: Principles and Practices, Sterling Publishers, New York 2. Carter, E and G. Lowman (1994): Ecotourism, John Wiley and Sons, New York, 3. Charles R. Goeldner, J.R. Brent Ritchie (2009); Tourism: Principles, Practices, Philosophies, John Wiley & Sons 4. Christopher M. Law (2009); Urban Tourism: The Visitor Economy and the Growth of Large Cities, Continuum Delhi 5. Kamra K.K. and Mohinder Chand (2007): Basics of Tourism: Theory, Operation and Practice, Kanishka Publishers New Delhi 6. Ministry of Tourism (2011); Strategic Action Plan for Tourism in India, Ministry of Tourism, Government of India 7. Prasad, Sheela and Kapil Kumar Gavske (2016): „The New „Love“ Story of the Taj Mahal-Urban Planning in the Age of Heritage Tourism in Agra“, Economic and Political Weekly, LI(5): 40-48 8. Robinson, H.A. (1996): Geography of Tourism, Macdonald and Evans, London 9. Williams, Stephen (1998): Tourism Geography, Routledge, London
Learning Outcomes	• Enhancing student's learning on causes and consequences of Migration • Students will be able to develop the knowledge of trends and patterns of Internal Migration. • They will be learning the knowledge of International Migration and Diaspora. • Students should understand Indian diaspora and its characteristics.

Course Title	GGRMJ83I: Fluvial Geomorphology			
Category of Course	Major			
Credits & Hour of Teaching		Theory	Practical	Cumulative
	Credits	4		
	Hour of Teaching (Total)	60		
Course Objectives	• To introduce students the basics and concepts of drainage basin and geometry. • To describe and explain channel morphology and depositional environment. • To understand and comprehend larger river systems and their actions. • To train students on river management and restoration.			
Course Content	The course content is divided into 4 Units.			
Units	Course Content			Hr. of Teaching
I	Overview of River Processes and Hydraulics: Drainage basin evolution; divide, channel network, overview of open-channel flow & Geometry overland flow, nodes and their classification; erosion and transport in river channel, Types of flow.			15
II	Channel Morphology & Fluvial Depositional Environment: The bank-full channel: Hydraulic geometry, Flow regimes and dominant discharge; Aggradation in the fluvial system: aggradation as a normal phase of valley evolution, Climatic change, alluvial valleys, terrace, River long profiles: Graded streams, base level, downstream fining channel alluvium, bar, Multi-thread channels, avulsions, floodplains, Delta and Alluvial fans.			15
III	Large River in the Tropics: Channel characteristics, hydrology and morphology, recent problems of Amazon, Zambezi, Ganga-Brahmaputra system, Kosi.			15
IV	River Management & Restoration: River health assessment, framework and principles, indicators, management and operational issues, case studies, Impact of Climatic & anthropogenic change on river.			15

SEC-Skill Enhancement Course, AEC Ability Enhancement Course; VAC Value Added Course: MD-Multidisciplinary Course 1 Credit (Theory)=15 Hours; 1 Credit (Practical)=30 Hours

Texts/References	1. C.R.Thorne(Editor), Richard D.Hey(Editor), Malcolm D.Newson(Editor), Applied Fluvial Geomorphology for River Engineering and Management, by John Wiley and Sons, 1998. ISBN: 978-0-471-96968-6 2. David Knighton Fluvial Forms and Processes: A New Perspective, John Wiley and Sons, 1998. ISBN 13: 9780340663134 3. S.Lawrence Dingman, Fluvial Hydraulics, Oxford University Press, 2009. ISBN 978-0-19-517286-7. 4. Ro Charlton, Fundamentals of Fluvial Geomorphology, Routledge, 2007 5. King, C.A.M. Techniques in Geomorphology, Edward Arnold, 1967 6. Kirstie A. Fryirs, Gary J. Brierley, 2012 Geomorphologic Analysis of River Systems: An Approach to Reading the Landscape, Wiley-Blackwell 7. Miller, 1964 Fluvial Processes in Geomorphology, by Luna Bergere Leopold, Markley Gordon Wolman, John Preston 8. River morphology, 2006, R.J. Garde, New Age International (P) Limited, Publisher, New Delhi 9. Small, R.J. The Study of Landforms – A Textbook of Geomorphology, Cambridge University Press, 1970
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Learning Outcomes	- Enhancing student's learning on drainage basin evolution; divide, channel network. - Students will be able to develop the knowledge about hydraulic geometry, flow regimes and dominant discharge. - They will learn the knowledge of channel characteristics. - Students should understand river health assessment and impacts of climatic change.
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Course	B.Sc./B.A. Fourth Year: Semester-VIII			
Course Title	GGRMN81: Advance Surveying			
Category of Course	Minor			
Credits & Hour of Teaching		Theory	Practical	Cumulative
	Credits		2	
	Hour of Teaching (Total)		60	
Course Objectives	- To introduce student's survey instruments Theodolite, Dumpy Level, Total Station and working principles. - To train students on GPS survey: Elements and segments; GPS Satellites.			
Course Content	The course content is divided into 2 Units.			
Units	Course Content			Hr. of Teaching
I	Survey Instruments: Theodolite, Dumpy Level, Total Station and GPS; Fundamentals of Total Station: parts, characteristics, advantages and disadvantages; Working Principles: settings, measurement (angular and linear), open and closed traverse			30
II	GPSSurvey: Elements and segments; GPS Satellites: Global Navigation Satellite System (GNSS), Indian Regional Navigation Satellite System (IRNSS); Mobile Mapping; location-based survey (LBS).			30

SEC-Skill Enhancement Course, AEC Ability Enhancement Course; VAC Value Added

Course: MD-Multidisciplinary Course 1 Credit (Theory)=15 Hours; 1 Credit (Practical)=30 Hours

Texts/References	- Rueger, J.M. Electronic Distance Measurement, Springer-Verlag, Berlin, 4th Edition, 1996. - Satheesh Gopi, R. Sathish Kumar, N. madhu, " Advanced Surveying, Total Station GPS and Remote Sensing " Pearson Education, 2nd Edition, 2017 R. Subramanian, Surveying and Levelling, Oxford University Press, Second Edition, 2012. - Laurila, S.H. Electronic Surveying in Practice, John Wiley and Sons Inc, 1983. - Guocheng Xu, GPS Theory, Algorithms and Applications, Springer-Verlag, Berlin, 3rd Edition, 2016. - Alfred Leick, GPS satellite surveying, John Wiley and Sons Inc., 4th Edition, 2015. - Seeber G, Satellite Geodesy, Walter De Gruyter, Berlin, 2nd Edition, 2003
Learning Outcomes	- Enhancing student's learning on survey instruments- parts, characteristics, advantages and disadvantages - Students should understand GPS survey and its importance.

Course	B.Sc./B.A.FourthYear:Semester-VIII			
CourseTitle	GGRMN82:Environmental Quality Analysis			
Category of Course	Minor			
Credits&Hourof Teaching		Theory	Practical	Cumulative
	Credits		2	
	Hour of Teaching (Total)		60	
CourseObjectives	- To introduce students soil and water quality assessment. - To train students on air, noise and solid waste assessment.			
CourseContent	The course content is divided into 2 Units.			
Units	CourseContent			Hr.of Teaching
I	Soil and water analysis: Soil health kundali: Calcium, magnesium, nitrogen, oxygen, humus, phosphorus, alkalinity and salinity, moisture (wet and dry), Water quality analysis: temperature, pH, DO, BOD and Water Quality Index (surface and groundwater), Point and non-point sources of pollution.			30
II	Air and Noise analysis and Solid waste: Air quality analysis: SO ₂ , NO ₂ , CO ₂ , PM 2.5, PM 10, Air Quality Index; Noise levels: Indoor and outdoor (Peak and non-peak hours of day and night); Measuring EMR by electric, magnetic, ultrasonic equipment; Solid waste: Environmental Impact Assessment of waste disposal and processing site.			30

SEC-Skill Enhancement Course, AEC Ability Enhancement Course; VAC Value Added Course: MD-Multidisciplinary Course 1 Credit (Theory)=15 Hours; 1 Credit (Practical)=30 Hours

Texts/References	- Rathi, A. Kumar, (2021) Handbook of Environmental impact assessment: concept and practice, Cambridge scholar publishing, New Castle, UK. - Rump, H. Hand Krist, H. (Eds) (1992): Laboratory Manual for the Examination of Water, Waste Water and Soil, Weinheim: VCH Publishers Ins, Germany. - U.S. Environmental Protection Agency, (2017): Quality Assurance Handbook for Air Pollution Measurement Systems Volume II, Office of Air Quality Planning and Standards, Air Quality Assessment Division, North Carolina, USA
Learning Outcomes	- Enhancing student's learning on soil and water analysis on select parameters. - Students to learn about environmental parameters of air, noise and solid waste.

For Honours Degree with Research

Course	B.Sc./B.A.FourthYear:Semester-VII			
CourseTitle	GGRMJ7R1:Research Methodology			
Category of Course	Major			
Credits&Hourof Teaching		Theory	Practical	Cumulative
	Credits	4		
	Hour of Teaching (Total)	60		
CourseObjectives	- To introduce the concept and significance of research geography. - To explain the philosophy and geography relations. - To introduce students about research process and design and data collection - To acquaint the students with data analysis, report writing and structure.			
CourseContent	The course content is divided into 4 Units.			
Units	CourseContent			Hr.of Teaching

I	Research in Geography: Concept and significance of research in geography; Production of geographical knowledge; Philosophy and Geography: Ontology, Epistemology, Methodology and Methods; Positivist and Non-Positivist Research	15
II	Research Process and Design : Research Process; Research designs: quantitative (experimental and survey), qualitative (ethnography, grounded theory, case study, phenomenological), mixed method (sequential, concurrent, and transformative); Ethics in Research	15
III	Data Collection Methods: Primary data collection: questionnaires, observation, key information interviews, in-depth interviews, focus group discussion; Sources of secondary data; Sampling and sample size determination; Quality control checks; Data cleaning.	15
IV	Data Analysis and Dissemination: Data analysis: Quantitative (variables, scales, univariate, bivariate, and multivariate analysis), Qualitative (thematic analysis); Writing reports and research papers: structure, referencing, plagiarism.	15

SEC-Skill Enhancement Course, AEC Ability Enhancement Course; VAC Value Added Course: MD-Multidisciplinary Course 1 Credit (Theory)=15 Hours; 1 Credit (Practical)=30 Hours

Texts/References	1. Ahuja, R. (2001): Research Methods. Rawat Publications, Jaipur and New Delhi. 2. Denzin, N.K. and Lincoln, Y.S., (eds.) (2000): Handbook of Qualitative Research. Thousand Oaks CA. Sage Publications. 3. Flowerdew, R. and Martin, D. (eds.) (1997): Methods in Human Geography. A Guide for Students Doing a Research Project. Longman, Harlow. 4. Kitchin, R. and Tate, N., (2001): Conducting Research into Human Geography: Theory, Methodology and Practice. Prentice-Hall, London. 5. Limb, M. (2001): Qualitative Methodologies for Geographers: Issue and Debates. Edward Arnold, London. 6. Mishra, H.N. & Singh, V.P. (2002): Research Methodology in Geography, Rawat Publications, Jaipur. 7. Neuman, Lawrence (2020): Social Research Methods: Qualitative and Quantitative Approach, Pearson 8. Rogers, A. and Viles, H.A. (2003): The Student's Companion to Geography. Blackwell Publishers, Oxford. Indian reprint available. 9. Silverman, D. (1993): Interpreting Qualitative Data: Methods for Analysing Talk, Text and Interaction. Sage Publications, London. 10. Tandon, B.C. (1997): Research Methodology in Social Sciences, Chaitanya Pub., Allahabad.
Learning Outcomes	• They would be able to enhance the learning on geographical research and its perspectives. • Students will be made aware about themes, approaches and recent developments. • They will be acquainted the knowledge of types of research designs and ethics. • Students should understand the data collection, analysis and presentation.

Course	B.Sc./B.A. Fourth Year: Semester-VII			
Course Title	GGRMJ7R2: Environmental Studies			
Category of Course	Major			
Credits & Hour of Teaching		Theory	Practical	Cumulative
	Credits	4		
	Hour of Teaching (Total)	60		
Course Objectives	• To introduce students to types and components of environment.			

	• Toexplainthedistributionandclassificationofnaturalresources. • Tointroducestudentsaboutenvironmentissuesandmonitoring. • Toacquaintthestudentswithenvironmentalconflictsandgeopolitics.	
CourseContent	Thecoursecontentisdividedinto4Units.	
Units	CourseContent	Hr.of Teaching
I	Environmentalconcepts: Meaning,typesandcomponentsoftheenvironment;ApproachestoEnvironmentalStudies;Ecosystem:componentsandclassification;Geo-biochemical cycles: carbon,nitrogen and phosphorus.	15
II	NaturalResourcesandBiodiversity: Distribution and classification of NaturalResources: Land, Water, Forest and Energy; Biodiversity: Extinction and conservation; Biomes of the world.	15
III	Environmental problems and management: EnvironmentalIssuesin Tropical, Temperate and PolarEcosystems;Pollution:air,water,soilandsolidwaste;Hazards :earthquakeandflood;Environmentalmonitoring;Environmentaleducation,Environmentalpoliciesandlegislations.	15
IV	TrendsinEnvironmentalstudies: Environmental conflicts; Environment and geopolitics; transnational movement; Environmental justice; Environmental geoinformatics: Environmental surveillance;Environmentalmapping.	15

SEC-Skill Enhancement Course, AEC Ability Enhancement Course; VAC Value Added Course; MD-Multidisciplinary Course 1 Credit (Theory)=15 Hours; 1 Credit (Practical)=30 Hours

Texts/References	- Anjune yulu, Y. (2004): Introduction to Environmental Science. B.S. Publications, Hyderabad. - Blaikie, P., Cannon, T. and Davis, I. (eds.) (2004): At Risk: Natural Hazards, Peoples Vulnerability and Disasters. Routledge, London. - Huggett, R. J. (1998): Fundamentals of Biogeography. Routledge, London. - Kayastha, S. L. and Kumra V. K. (1986): Environmental Studies. Tara Book Agency, Varanasi. - Kumaraswamy, K., Alagappa Moses, A. & Vasanthi, M. (2004) <i>Environmental Studies</i>, Bharathidasan University, Tiruchirappalli. - Nag, P., Kumra, V. K. and Singh, J. (1990): Geography and Environmental Issues at Local, Regional and National Levels. (in 3 vols.), Concept Publishing Company, New Delhi. - Odum, E. P. (1975): Ecology. Rowman and Littlefield, Lanham USA. - Rajagopalan, R. (2005): Environmental Studies: From Nrisisto Cure, Oxford University Press, New - Saxena, H. M., 2012: Environmental Studies, Rawat Publications, Jaipur. - UNEP, (2007): Global Environment Outlook: GEO4: Environment for Development, United Nations Environment Programme
Learning Outcomes	- Enhancing student's learning on types, components of environment and approaches. - To make students aware about distribution and biodiversity challenges. - Acquainted the knowledge of environmental issues, monitoring and policies. - Students should understand the environment geopolitics, justice and mapping.

Course	B.Sc./B.A. Fourth Year: Semester-VII			
Course Title	GGRMJ7R3: Geographical Field Training and Report Writing			
Category of Course	Major			
Credits & Hour of Teaching	Credits	Theory	Practical	Cumulative
			2	

	Hour of Teaching (Total)		60	
Course Objectives	• To introduce students, the meaning and objectives of geographical field training • To explain field as a laboratory of geography. • To understand various contents of field training. • To train students on field based report writing.			
Course Content	The course content is divided into 2 Units.			
Units	Course Content			Hr. of Teaching
I	Principles of Field Training: Meaning and objectives of field training; Field as a laboratory of geography; Contents of field training: physical, social, economic and cultural; Field training techniques; Field training-based report writing.			30
II	Field Visit: Students will visit any one of the following geographical regions of physical and human interest for field training: Himalayan Region, Vindhyan Plateau, Thar Desert and Aravali Hills, Coastal region, East and North-eastern India.			30

SEC-Skill Enhancement Course, AEC Ability Enhancement Course; VAC Value Added Course: MD-Multidisciplinary Course 1 Credit (Theory)=15 Hours; 1 Credit (Practical)=30 Hours

Texts/References	1. Archer, J.E. and Dalton, T.H. (1968): Field Work in Geography. William Clowes and Sons Ltd. London and BeM Jles. 2. Bolton, T. and Newbury, P.A. (1968): Geography through Fieldwork. Blandford Press, London. 3. Jones, P.A. (1968): Field Work in Geography. Longmans, Green and Company Ltd., London and Harlow. 4. Lousenbury, J.F. and Aldrich, F.T. (1986): Introduction to Geographic Field Methods and Techniques. Charles E. Merrill Publishing. Company, Columbus. 5. Pugh, J.C. (1975): Surveying for Field Scientists. Methuen and Company Ltd. London. 6. Parsons, Tony and Knight, Peter G. (2005): How to do your Dissertation in Geography and Related Disciplines. Routledge, London. 2nd Ed. 7. Kitchen, Rob and Tate, Nicholas J. (2009): Conducting Research into Human Geography: Theory, Methodology & Practice. Prentice Hall-Pearson, Harlow U.K. 2nd Ed. 8. Hay, Iain (ed.) (2005): Qualitative Research Methods in Human Geography. Oxford University Press, Melbourne. 2nd Ed. 9. Hay, Iain (ed.) (2004): Communicating in Geography and the Environmental Sciences. Oxford University Press, Melbourne. 2nd Ed. 10. Stoddard, Robert H. (1982): Field Techniques and Research Methods in Geography. Kendall/Hunt Pub. Dubuque IO.
Learning Outcomes	• Enhancing student's learning on importance of fielding geographical knowledge. • To make students aware about various important aspects of field phenomenon. • Acquainted the knowledge of major techniques. • Students should understand the field training-based report writing..

Course	B.Sc./B.A. Fourth Year: Semester-VII			
Course Title	GGRMJ7R4A: Advanced Geographical Thought			
Category of Course	Major			
Credits & Hour of Teaching		Theory	Practical	Cumulative
	Credits	4		
	Hour of Teaching (Total)	60		
Course Objectives	• To introduce students the recent nature of geographical knowledge. • To explain new environmentalism and major approaches in geography. • To understand various paradigms and contributions of geographers.			

	• To train students on contribution of Indian geographers and scholars.	
Course Content	The course content is divided into 4 Units.	
Units	Course Content	Hr. of Teaching
I	Basic Concepts: Nature of geographical knowledge; Man-environment interaction: new environmentalism; Concepts: space, place, environment, time and spatial organisation; Culture and cultural all and scape.	15
II	Modern Approaches: Quantitative geography; Humanistic geography: phenomenology and existentialism; Contributions of Carl Sauer and Yi-Fu Tuan.	15
III	Contemporary Trends: Qualitative paradigm; post-modernism and post-structuralism; Social theory and De/reconstruction. Welfare/Marxist/ Radical geography: contributions of David Harvey, Feminist geography.	15
IV	Trends of Indian Geography: Colonial and Postcolonial geography; Contributions of Indian geographers; Impact of thoughts of Gandhi, Nehru and Ambedkar on Indian geography; Indian geography: issues and challenges.	15

SEC-Skill Enhancement Course, AEC Ability Enhancement Course; VAC Value Added Course: MD-Multidisciplinary Course 1 Credit (Theory)=15 Hours; 1 Credit (Practical)=30 Hours

Texts/References	1. Daniels, P., Bradshaw, M., Shaw, D. and Sidaway, J. (2000): An Introduction to Human Geography. Issues for the 21st Century. Prentice Hall, London. 2. Dear, M. J. and Flusty, S. (2002): The Spaces of Postmodernity: Readings in Human Geography. Blackwell Publishers, Oxford. 3. Dikshit, R. D. (2004): Geographical Thought. A Critical History of Ideas. Prentice-Hall India, New Delhi. 4. Harvey, M. E. and Holly, P. B. (2002): Themes in Geographic Thought. Rawat Publications, New Delhi. 5. Hubbard, P., Kitchin, R., Bartley, B. and Fuller, D. (2002): Thinking Geographically: Space, Theory and Contemporary Human Geography. Continuum, London. 6. Johnston, R. J. and Sidaway, J. D. (2004): Geography and Geographers. 6th edition, Edward Arnold, London. 7. Kapur, A., (ed.) (2001): Indian Geography-Voice of Concern. Concept Publishing Company, New Delhi. 8. Martin, G., (2005): All Possible Worlds. A History of Geographical Ideas. 4th edition, Oxford University Press, New York. 9. Mathews, J. A., and Herbert, D. T. (eds.) (2004): Unifying Geography. Common Heritage, Shared Future. Routledge, London. 10. Soja, E. (1989): Post-modern Geographies. Verso Press, London. Reprinted 1997: Rawat Publications, Jaipur and New Delhi 11. Taylor, G. (ed.) (1953): Geography in the Twentieth Century. Mathuen and Company Ltd. and Company, London. 12. Tuan, Yi-Fu (1977): Space and Place. The Perspective of Experience. Edward Arnold, London
Learning Outcomes	• Enhancing student's learning on changing nature geographical knowledge. • To make students aware about various modern approaches. • Acquainted the knowledge of contemporary trends and contributions. • Students should understand the India's role in postcolonial geography.

Course	B.Sc./B.A. Fourth Year: Semester-VII			
Course Title	GGRMJ7R4B: Advances in Population Studies			
Category of Course	Major			
Credits & Hour of Teaching		Theory	Practical	Cumulative
	Credits	4		
	Hour of			

	Teaching (Total)	60		
Course Objectives	• To introduce students the meaning, scope and approaches to Population Studies. • To explain the factors, determinants and patterns & theories of population. • To understand composition, structure and factors of change in population. • To train students on population policies and contemporary issues.			
Course Content	The course content is divided into 4 Units.			
Units	Course Content		Hr. of Teaching	
I	Basics of Population Studies: Meaning, Scope and approaches to Population studies; Sources of Population data with special reference to India; Demographic Dividend; Recent trends in population studies.		15	
II	Population Distribution, Density and Growth: Factors, Determinants and Patterns; Theories of Population Growth: Malthusian theory and Demographic Transition Theory; Concept of Under, Over and Optimum Population; Population Resource Region.		15	
III	Population Composition and Migration: Age-Sex, Literacy, Rural-Urban, Occupation structure; Fertility; Mortality; Migration: Concept, Types, Determinants and Theories;		15	
IV	Population policies and issues: Population Policies in Developed and Developing countries (A Case of India); Contemporary Issues in Population Geography- Ageing Population, Sex Ratio, Diseases, Epidemics and Population Problems; Population and Environment Interface; Population and Development: Human Development Index; New Research areas in Population Geography;		15	

SEC-Skill Enhancement Course, AEC Ability Enhancement Course; VAC Value Added

Course: MD-Multidisciplinary Course 1 Credit (Theory)=15 Hours; 1 Credit (Practical)=30 Hours

Texts/References	1. Chandna, R.C. (2010) Population Geography, Kalyani Publisher, New Delhi 2. Hassan, M.I. (2020) Population Geography: A Systematic Exploration, Routledge India Pvt. Ltd, New Delhi 3. Tiwari, R.C. (2007) Geography of India. Prayag Pustak Bhawan, Allahabad 4. Barrett H.R., 1995: Population Geography, Oliver and Boyd. 5. Bhende A. and Kanitkar T., 2000: Principles of Population Studies, Himalaya Publishing House. 6. Chandna R.C. and Sidhu M.S., 1980: An Introduction to Population Geography, Kalyani Publishers 7. Jones, H.R., 2000: Population Geography, 3rd ed. Paul Chapman, London. 8. Lutz W., Warren C.S. and Scherbov S., 2004: The End of the World Population Growth in the 21st Century, Earthscan 9. Newbold K.B., 2009: Population Geography: Tools and Issues, Rowman and Littlefield Publishers 10. Pacione M., 1986: Population Geography: Progress and Prospect, Taylor and Francis. 11. Wilson M.G.A., 1968: Population Geography, Nelson.
Learning Outcomes	• Enhancing student's learning on basics of population Geography. • To make students aware about various factors affecting population growth. Acquainted the knowledge of composition and structure of population. • Students should understand population issues and policies.

Course	B.Sc./B.A.FourthYear:Semester-VII			
CourseTitle	GGRMJ7R4C:Urban Geography			
Category of Course	Major			
Credits&Hour of Teaching		Theory	Practical	Cumulative
	Credits	4		
	Hour of Teaching (Total)	60		
CourseObjectives	- To introduce students the nature, scope and recent trends in urban geography. - To explain the origin and evolution of urban settlements and their characteristics - To understand the city size and urban structure models. - To train students on urban growth and contemporary issues in planning.			
CourseContent	The course content is divided into 4 Units.			
Units	CourseContent			Hr. of Teaching
I	Bases: Nature, scope and recent trends in urban geography; Urban population: characteristics, processes and trends; Methodology in urban studies; Origin and evolution of urban settlements: bazaar-based and colonial based; distribution of urban centres.			15
II	Characteristics: Characteristics of cities in different historical periods: pre-industrial, industrial and post-industrial; Functional classification of urban centres and urban transportation; National Urban Information System.			15
III	Spatiality and Models: Size and spacing of cities: rank-size rule; law of the primate city; Urban hierarchies; Central place theory (Christaller and Loch); Urban land use and functional morphology: functional areas and peri-urban areas; Theories of urban structure: Burgess, Hoyt, Harris and Ullman, Mann, and White.			15
IV	Issues and Planning: Urban sprawl; urban problems: environmental, poverty, inequality, slums, transportation, housing, segregation and crime; Planned cities: Chandigarh and Jaipur; smart city in India; National Urban Policy; Master Plan: A case study of Varanasi.			15

SEC-Skill Enhancement Course, AEC Ability Enhancement Course; VAC Value Added Course; MD-Multidisciplinary Course 1 Credit (Theory)=15 Hours; 1 Credit (Practical)=30 Hours

Texts/References	1. Bridge, B. and Watson, S. (Eds.) (2000): A Companion to the City. Blackwell, Oxford. 2. Carter, H. (1995): The Study of Urban Geography. 4th ed. Reprinted in 2000 by Rawat Publications, Jaipur and New Delhi. 3. Dutt, A. Allen, K. Nobel, G., Venugopal G. and Subbiah S. (eds.) (2003): Challenges to Asian Urbanisation in the 21st Century. Kluwer Academic Publishers, Dordrecht and London. 4. Hall, T. (2001): Urban Geography. 2nd edition. Routledge London. 5. Mayer, H. and Cohn, C. F. (1959): Readings in Urban Geography, University of Chicago Press, Chicago 6. Pacione, M. (2005): Urban Geography: A Global Perspective. Routledge, London and New York. 7. Ramachandran, R., (1991) Urbanization and Urban Systems in India. Oxford University Press, Delhi. 8. Singh, R. L. (1955): Banaras. A Study in Urban Geography. Nand Kishore and Brothers Banaras 9. Singh, H. H. (1972): Kanpur: A Study in Urban Geography, Indrasini Publications, Varanasi 10. Stanley, B., Jack, W. and Donald, Z. (eds.) (2003): Cities of the World. Rowman and Littlefield, New York and Oxford. 11. Tripathi, V. K., Jha, D. K. and Harshwardhan R. (eds.) (2018): Advances in Urban Studies in India. R. K. Books, New Delhi.
Learning Outcomes	Enhancing student's learning on bases and growth of Urban Geography.

	• To make students aware about various factors affecting urban population. • Acquainted the knowledge of composition, urban structure and functions. • Students should understand contemporary urban population issues, policies and planning.
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Course	B.Sc./B.A.Fourth Year: Semester-VII			
Course Title	GGRMJ7R4D: Regional Planning			
Category of Course	Major			
Credits & Hour of Teaching		Theory	Practical	Cumulative
	Credits	4		
	Hour of Teaching (Total)	60		
Course Objectives	• To introduce students the concept, nature, and scope of regional planning. • To describe theories, approaches and geographical issues of regional planning • To understand the evolution and growth of regional planning in India • To train students on contemporary regional development issues and planning.			
Course Content	The course content is divided into 4 Units.			
Units	Course Content			Hr. of Teaching
I	Conceptual Background: Concept, Nature and scope of regional planning; Theories and approaches of regional planning, Concept of Planning region, Planning regions of India; Decentralized planning			15
II	Regional Development: Regional imbalances and inequalities in India; Regional planning and national development; Economic development and regional development; Role of agriculture, industry and infrastructure in regional development.			15
III	Regional Planning: Evolution of Regional Planning in India; Regional planning and National development; Planning for regional development, Integrated regional planning, Multi-level planning, Survey for planning; diagnostic, techno-economic and regional			15
IV	Planning and Policy in India: Regional development strategy in India; Changing paradigm of planning in India. National and state level planning, Local level planning; National Capital Region; Planning for Eastern Uttar Pradesh.			15

SEC-Skill Enhancement Course, AEC Ability Enhancement Course; VAC Value Added

Course: MD-Multidisciplinary Course 1 Credit (Theory)=15 Hours; 1 Credit (Practical)=30 Hours

Texts/References	1. Chandna, R.C. (2000): Regional Planning: A Comprehensive Text. Kalyani Publishers., New Delhi. 2. Chaudhuri, J.R. (2001): An Introduction to Development and Regional Planning with special reference to India. Orient Longman, Hyderabad. 3. Friedmann, J. and Alonso, W. (ed.) (1973): Regional Development and Planning. The MIT Press, Mass. 4. Kuklinski, A.R. (1972): Growth Poles and Growth Centres in Regional Planning. Mouton and Co., Paris. 5. Misra, R.P. (ed.) (1992): Regional Planning: Concepts, Techniques, Policies and Case Studies. 2nd edition. Concept Publishing Company., New Delhi. 6. Pathak, C.R. (2003): Spatial Structure and Processes of Development in India. Regional Science Association., Kolkata. 7. Sanyal, B.M. (2001): Decentralized Planning: Themes and Issues. Concept Publishing Company, New Delhi. 8. Sharma, P. V., Rao, V. L., and Pathak, C. R. (eds.) (2000): Sustainable Regional Development (with special reference to Andhra Pradesh). Regional Science Association, Kolkata and School of Economics, Andhra University, Vishakhapatnam. 9. Sundaram, K.V. (1997): Decentralized Multilevel Planning: Principles and Practice
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	ce(AsianandAfricanExperiences).Concept PublishingCompany,NewDelhi. 10. Sundaram,K. V.(2004):TheTroddenPath:EssaysonRegionalandMicroLevelPlanning.AnaunyaPublications.,NewDelhi.
Learning Outcomes	• Enhancingstudent'slearningonmajorfundamentalaspectsofregionalplanning. • Tomakestudentsawareaboutvariousfactorsandissuesofregionalimbalances. • Acquaintedtheknowledgeofevolutionofregional planninginIndia. • Studentshouldunderstandcontemporaryregionaldevelopmentstrategy.